
Emergent Communication Under Misinformation

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Abstract

Social interactions are characterized by both adversarial and cooperative aspects. Communications between agents may also involve adversarially motivated actors. Messages can pass through intermediaries with malicious intent before reaching the intended receiver. These actors may modify the message to induce misunderstanding from the receiver while preserving the overall characteristics of the message. This form of misinformation is prevalent in real-world communications and may affect the dynamics under which communication protocols are developed. However, this aspect of social interaction is relatively underexplored in many emergent communication studies that aim to understand the environmental factors behind the emergence of languages' characteristics. This work explores how misinformation affects language emergence with a focus on compositionality. We design a communication game containing a malign intermediary between the sender and receiver. We find that risks of malign misrepresentation promote the emergence of compositional languages in simulations of communicative agents. Furthermore, we observe that adaptability of malign intermediaries is a crucial factor in forming a consistent pressure toward compositionality and that partial misinformation, in which the intermediary targets only a subset of attributes, can also induce compositionality.

1. Introduction

Human languages exhibit a diverse set of characteristics which were accumulated over an extended period of evolution. Many of these properties reflect environmental factors in which the languages are learned and used, including the limited capacity of human cognition (Christiansen &

Chater, 2016), and types of social interactions (Raviv et al., 2019). Determining these environmental factors that shape the evolutionary process of languages enables us to better understand each property's role in human communication, and holds potential to shed light on how humans as well as machines can more effectively learn to utilize such structures in communication (Zheng et al., 2024). However, the evolutionary process of spoken language is often hard to trace, and there is a limit to asserting a causal link from static historical evidence as it does not permit intervention (Scott-Phillips & Kirby, 2010).

Emergent communication provides a controlled environment for identifying the core environmental factors behind the emergence of a given characteristic of language via simulations of communicative agents. It allows designing specific environmental pressures, and this approach has been successful in determining the contributing factors for the emergence of various linguistic structures (Lian et al., 2023; Ueda et al., 2023; Ueda & Washio, 2021). Recent developments in deep learning have renewed interest in the area (Lazaridou et al., 2018), and its findings on linguistic structures inspire the development of practical applications in the broader machine learning community (Elberg et al., 2025; Zheng et al., 2024; Ren et al., 2023).

Compositionality, the ability to express complex meanings by combining simpler parts, underwrites the impressive productivity of human language. This characteristic allows humans to communicate a seemingly unlimited number of concepts with a finite set of symbols by combining them with a systematic rule structure. This property is often considered a hallmark of human language, and it supports efficient learning (Smith et al., 2003; Ren et al., 2020; Li & Bowling, 2019), systematic generalization to unseen combinations (Chaabouni et al., 2020; Ren et al., 2023; Rita et al., 2022b), and reuse of learned components (Jiang et al., 2019).

Accordingly, understanding how and when compositionality arises is actively investigated in the field, and various conditions have been identified as key factors that shape its emergence. These aspects include channel reliability (Kuciński et al., 2021), types of social structures of communicating agents (Michel et al., 2023; Yu et al., 2022), as well as task difficulty (Chaabouni et al., 2022). However, most

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prior work assumes either a perfectly clean communication channel (Chaabouni et al., 2020; Guo et al., 2023; Rita et al., 2022b) or undirected random noisy channels (Kuciński et al., 2021; Nikolaus, 2024; Klamra et al., 2025; Ueda & Washio, 2021; Foerster et al., 2016). Yet, real-world communications contain risks of message modification from actors with malicious intent. A sender’s message may be relayed by an intermediary before arriving at the intended recipient. This agent could modify the message to induce misunderstanding while preserving the overall appearance of the message. Although misinformation by malign intermediaries is commonly observed in everyday communications, its effects on language emergence remain underexplored.

To investigate the effects of misinformation on language emergence, we design a communication game under misinformation by introducing a malign intermediary between the sender and receiver. This agent, with some probability, intercepts the sender’s message and modifies a limited number of symbols in the message with the intent to cause misunderstanding on the receiver’s side. Sender and receiver are rewarded for successful communication, while the intermediary is rewarded for inducing errors under the allowed modification budget.

We hypothesize that risks of misinformation can apply a stronger pressure toward compositionality as holistic expressions that do not represent meaning in a structured distributional manner are more severely penalized. We theoretically analyze how misinformation can induce preference toward compositional protocols and find that the empirical results confirm the hypothesis.

We then explore the mechanisms that govern how misinformation promotes compositionality, and find that the malign intermediary’s adaptability is a crucial factor. As the sender and receiver learn to communicate under misinformation, their protocols change to circumvent the intermediary. An intermediary that fails to continuously track these changes may be taken advantage of by the sender and receiver, thereby exerting a weaker and less consistent pressure. Our theoretical analysis demonstrates how such an intermediary may hinder the emergence of compositionality. We hypothesize that introduction of new malign intermediaries can more effectively track the changes. Inspired by generational transmission of language in the iterated learning framework (Smith et al., 2003), we simulate new generations of intermediaries during language development and find that introduction of new intermediaries can exert an adaptive and consistent pressure toward the emergence of compositionality.

To better reflect the realistic communication environment, we introduce partial misinformation where the intermediary is only interested in inducing misunderstanding on a subset of attributes contained in a message and is indifferent to

other attributes. The experimental results indicate that partial misinformation can also induce the emergence of compositionality, suggesting that the intermediary’s interests need not encompass the entirety of information represented in a message. Moreover, in some cases, we observe that it promotes higher degrees of compositionality compared to the all-attribute misinformation setup.

2. Related work

Emergent communication Emergent communication is widely used to study the relevant factors that influence the emergence of linguistic structures. These properties include word boundary formation (Ueda et al., 2023; Harris, 1955), negation (Todd et al., 2020), case marking (Lian et al., 2023), word-order biases (Chaabouni et al., 2019), and Zipf’s law (Zipf, 1949; Ueda & Washio, 2021). One of the central themes of the field is understanding the conditions that promote the emergence of compositional languages. Various environmental conditions such as population size and structure (Rita et al., 2022a; Michel et al., 2023; Chaabouni et al., 2022; Kim & Oh, 2021), capacities of channels and agents (Chaabouni et al., 2020; Resnick et al., 2020), task difficulty (Chaabouni et al., 2022), and types of tasks (Zion et al., 2024) have been identified as influential factors. However, most of these works assume a clean communication channel, even when they model adversarial aspects of social interactions (Yu et al., 2022; Blumenkamp & Prorok, 2020; Kharitonov et al., 2020). This work addresses this gap by investigating the role of misinformation in language emergence.

Relation to the broader machine learning field The findings of emergent communication studies also inform developments in the broader machine learning field. The identified contributory pressures for the emergence of compositionality are employed to learn disentangled representations (Xu et al., 2022; Elberg et al., 2025). Exploration of how such pressures can help the models learn the underlying rule structures of the data, thereby improving the compositional generalization capabilities (Ren et al., 2023; Zheng et al., 2024) or preventing language drift (Noukhovitch et al., 2023), is also an active area of research. The language-like properties of emergent languages are leveraged to alleviate the data scarcity issue in various domains such as embodied control (Mu et al., 2023), language modeling (Yao et al., 2022; Steinert-Threlkeld et al., 2022), and machine translation (Li et al., 2020; Downey et al., 2023).

Reliability of communication channels Studies that explore how channel reliability can promote structured meaning representations (Nikolaus, 2024; Ueda & Washio, 2021; Klamra et al., 2025; Foerster et al., 2016) are closely related to this work. Nowak & Krakauer (1999) suggest that

formulating words with combinations of a set of easily distinguishable sounds, rather than using a single sound slot, is to avoid misunderstanding that may stem from packing too many meanings into a limited set of distinguishable sounds. They employ evolutionary game theory to analyze how compositionality may be a preferred solution in this environment and show that the use of compositional languages is an evolutionarily stable strategy. Kuciński et al. (2021) extend this idea and explore the effects of random channel noise on the emergence of compositionality. They theoretically prove that compositional languages are optimal under random noise and empirically demonstrate that noisy channels can facilitate the emergence of compositional languages.

Introduction of new generations in language evolution

How new generations learn to use the previous generation’s language is another important aspect affecting the emergence of compositionality. The iterated learning framework (Smith et al., 2003) is widely used to study this phenomenon. Numerous studies show that the pressure to be easy-to-learn by new generations forces the emergence of compositionality (Ren et al., 2020; Guo et al., 2019; Li & Bowling, 2019; Chaabouni et al., 2020). These findings inspire developments on how general neural networks can acquire a structured understanding of the training samples (Ren et al., 2023; Zheng et al., 2024). Ren et al. (2024) show that this framework can be used to analyze how biases in large language models can be amplified over their evolution. This work shares similarities with this line of research in that new generations of language users are modeled, but differs in terms of which agents are introduced. We analyze how new generations of malign intermediaries affect the emergence of compositionality.

3. Communication game under misinformation

We investigate the effects of misinformation in language emergence with agents engaged in a Lewis-style signaling game (Lewis, 1969). The sender π_ϕ observes an object $x \in \mathcal{X}^N$. An object is characterized by N attributes indexed by $\mathcal{A} = \{1, \dots, N\}$ and each attribute takes a value in $\mathcal{X}$. The sender outputs a message $m \sim \pi_\phi(\cdot | x)$ describing the object x and sends the message to the receiver π_θ . A message consists of a fixed length of L symbols, and each symbol belongs to the alphabet $\mathcal{V}$. With probability p , the receiver π_θ receives the original message and predicts the object x by $\hat{x} \sim \pi_\theta(\cdot | m)$. With probability $1 - p$, the message is sent to the malign intermediary π_ψ interested in a set of attributes indexed by $\mathcal{S} \subseteq \mathcal{A}$. The malign intermediary observes the message m and the object x and modifies $k \leq L$ symbols within the message with the intent of inducing an incorrect prediction from the receiver π_θ as $m' \sim \pi_\psi(\cdot | m, x)$. The modified message is relayed to the receiver

and the receiver produces its prediction for the object x from the modified message m' as $\hat{x} \sim \pi_\theta(\cdot | m')$. The sender and receiver are rewarded if the receiver outputs the correct prediction for the object x either from m or m' by $\frac{1}{N} \sum_{n=1}^N \mathbb{1}(\hat{x}_n = x_n)$. The intermediary is rewarded when the receiver outputs an incorrect prediction of the object x on the attributes indexed by $\mathcal{S}$ from the modified message m' as $\frac{1}{|\mathcal{S}|} \sum_{n \in \mathcal{S}} \mathbb{1}(\hat{x}_n \neq x_n)$. $\mathbb{1}$ and x_n denote the indicator function and the value of the n -th attribute of the object x .

4. Experimental setup

Game setup We set the probability that an original message is directly passed to the receiver, p , to 0.5. We vary the value of k , the number of symbols modified by the malign intermediary, from 1 to 4 and observe its effect on the language structure. The number of modified symbols is kept exactly at k by reverting back (or modifying) randomly chosen modified (or unmodified) symbols if the number of modified symbols does not match k in the modified message (see Appendices F.1 and F.2 for a discussion of the validity of this post hoc correction mechanism). We compare directed misinformation of malign intermediaries with undirected random noise, where randomly chosen k symbols are modified to take different symbols each with probability $\frac{1}{|\mathcal{V}| - 1}$. A clean communication channel setup is also compared, where no modifications are made to the messages. The message length, L , and the alphabet size, $|\mathcal{V}|$, are both set to 10. All agents are GRUs (Cho et al., 2014) of one recurrent layer with hidden size 128. See Appendix B for a full description.

Dataset We utilize a synthetic attribute-value dataset where each attribute is represented by a one-hot vector and concatenated with the other one-hot vector attributes to form an object. We set the number of attributes to $N = 5$, and the number of values per attribute to $|\mathcal{X}| = 10$. We take 5% of the object combinations as the train set, 10% as the validation set, and use the remaining objects as the test set.

Optimization The receiver is trained to minimize the average cross entropy loss over its predicted attribute distributions. This is written as $-\frac{1}{N} \sum_{n=1}^N \log \pi_\theta(x_n | m)$, where x_n denotes the value of the n -th attribute of object x and m is the received message. This loss is directly backpropagated to the receiver. The sender and intermediary are optimized with the REINFORCE algorithm (Williams, 1992). The negative of the receiver’s loss is used as the reward for the sender. The intermediary’s reward is given by the corresponding cross entropy loss over the targeted attributes $\mathcal{S}$, which is written as $-\frac{1}{|\mathcal{S}|} \sum_{n \in \mathcal{S}} \log \pi_\theta(x_n | m)$. We add the average negative entropy of the symbol distributions as an auxiliary loss to encourage exploration for the sender and malign intermediary. See Appendix C for a full description

and Appendix G for pseudocode.

Reporting We report average results over 20 random seeds. Error bars represent 95% confidence intervals. Following Chaabouni et al. (2020), we analyze properties of languages at the end of training. Languages are evaluated in the absence of any modifications. We assess statistical significance using one-tailed Welch’s t -tests with $p < 0.05$.

5. Compositionality measures

We employ the following compositionality measures to analyze the emergent languages induced from the experiments.

Topographic similarity (TopSim) Topographic similarity (Brighton & Kirby, 2006) is Spearman’s rank correlation between object-space distances D_{obj} and message-space distances D_{msg} across uniformly sampled object-message pairs. D_{obj} and D_{msg} refer to the distance measures in the object and message spaces, which are written as $D_{\text{obj}} : \mathcal{X}^N \times \mathcal{X}^N \rightarrow \mathbb{R}^+$ and $D_{\text{msg}} : \mathcal{V}^L \times \mathcal{V}^L \rightarrow \mathbb{R}^+$. Larger values indicate stronger preservation of object-space neighborhood structure in the corresponding message space. For example, mappings like "e-mail" and "voicemail" would induce a high score, whereas "voicemail" and "blackmail" would not, as changing "voice" to "black" yields a completely different meaning. We use Hamming distance for D_{obj} and Levenshtein distance (Levenshtein, 1965) for D_{msg} .

Positional disentanglement (PosDis) Positional disentanglement (Chaabouni et al., 2020) measures how much a single position uniquely represents information about an attribute. Let m_i represent the symbol at the i -th position in message m . The entropy of the i -th position is written as $\mathcal{H}(m_i) = -\sum_{v \in \mathcal{V}} \mathbb{P}(m_i = v) \log \mathbb{P}(m_i = v)$, where $\mathbb{P}(m_i = v)$ denotes the probability that the i -th position contains the symbol v . Let a_1^i be the attribute that maximizes the mutual information with m_i . More specifically, $a_1^i = \arg \max_a \mathcal{I}(m_i; a)$, where $\mathcal{I}(\cdot; \cdot)$ denotes mutual information. Let a_2^i be the attribute that maximizes the mutual information with m_i among the rest of the attributes, i.e., $a_2^i = \arg \max_{a \neq a_1^i} \mathcal{I}(m_i; a)$. PosDis is written as $\frac{1}{L} \sum_{i=1}^L \frac{\mathcal{I}(m_i; a_1^i) - \mathcal{I}(m_i; a_2^i)}{\mathcal{H}(m_i)}$.

Reverse positional disentanglement (RevPosDis) We devise reverse positional disentanglement as a complementary metric to PosDis to measure the degree to which an attribute is exclusively represented by a single position. Let a_j denote the j -th attribute. Let m_1^j be the position that maximizes the mutual information with a_j , i.e., $m_1^j = \arg \max_{i \in \{1, \dots, L\}} \mathcal{I}(m_i; a_j)$, and let m_2^j be the position with the second-highest mutual information, i.e.,

$$m_2^j = \arg \max_{i \neq m_1^j} \mathcal{I}(m_i; a_j). \text{ RevPosDis is written as } \frac{1}{N} \sum_{j=1}^N \frac{\mathcal{I}(a_j; m_1^j) - \mathcal{I}(a_j; m_2^j)}{\mathcal{H}(a_j)}, \text{ where } \mathcal{H}(a_j) \text{ denotes the entropy of attribute } a_j.$$

Bag-of-symbols disentanglement (BosDis) Bag-of-symbols disentanglement (Chaabouni et al., 2020) measures the degree to which occurrences of an individual symbol exclusively convey information about an attribute. Let n_i be the number of times the i -th symbol from alphabet $\mathcal{V}$ appears in a message. Let $\mathcal{H}(n_i)$ denote the entropy of n_i , i.e., $\mathcal{H}(n_i) = -\sum_{c=0}^L \mathbb{P}(n_i = c) \log \mathbb{P}(n_i = c)$. Using the same notations as in PosDis, BosDis is written as $\frac{1}{|\mathcal{V}|} \sum_{i=1}^{|\mathcal{V}|} \frac{\mathcal{I}(n_i; a_1^i) - \mathcal{I}(n_i; a_2^i)}{\mathcal{H}(n_i)}$.

Compositional generalization We measure the generalization ability of languages with accuracy on the test set containing unseen attribute combinations.

6. Theoretical analysis setup

We follow the same approach as in Kuciński et al. (2021) and consider one-to-one languages, $\ell : \mathcal{X}^N \rightarrow \mathcal{V}^L$, which can be thought of as deterministic sender, $\pi_\phi(x) = \ell(x)$, and receiver, $\pi_\theta(m) = \ell^{-1}(m)$, pairs, and the dimension alignment between object and message spaces, i.e., the number of values per attribute is equal to the alphabet size ($|\mathcal{X}| = |\mathcal{V}|$), and the message length is equal to the number of attributes ($N = L$).

Definition 6.1 (One-to-one languages). A language ℓ is a one-to-one language if for every $x \in \mathcal{X}^N$ there exists a unique $m \in \mathcal{V}^L$ such that $\ell(x) = m$ and vice versa.

We also use the same definitions of compositionality, random noise corruption, along with the payoff function, as in Kuciński et al. (2021).

Definition 6.2 (Compositionality of one-to-one languages). A one-to-one language ℓ is compositional if and only if for every $x_1, x_2 \in \mathcal{X}^N$, $\rho(x_1, x_2) = 1$ implies $\rho(\ell(x_1), \ell(x_2)) = 1$, where ρ is the Hamming distance.

This definition of compositionality captures the notion that local changes in objects should remain local in the corresponding messages. When object and message dimensions are aligned, this is equivalent to requiring that similar objects be mapped to similar messages. See Lemma A.1 in Appendix A for details.

Definition 6.3 (Random noise). Random noise, Q_ϵ , modifies each symbol in a message with probability ϵ to another symbol with equal probability and preserves it with probability $1 - \epsilon$,

$$Q_\epsilon(m' | m) = (1 - \epsilon)^{L - \rho(m', m)} \left(\frac{\epsilon}{|\mathcal{V}| - 1} \right)^{\rho(m', m)}.$$

Definition 6.4 (Expected payoff). Expected payoff of language ℓ under corruption scheme Q is defined as

$$J(\ell, Q) := \mathbb{E}_{x \sim \text{Unif}(\mathcal{X}^N)} \mathbb{E}_{m' \sim Q(\cdot | m)} \frac{1}{N} \sum_{n=1}^N \mathbb{1}[x'_n = x_n],$$

where $m = \ell(x)$ and $x' = \ell^{-1}(m')$.

We define the set of admissible corruption schemes as below.

Definition 6.5 (Admissible corruption schemes). For $k \in [0, L]$, let $\mathcal{Q}_k$ denote the set of corruption schemes Q such that $\mathbb{E}_{m' \sim Q(\cdot | m)} \rho(m', m) = k$ for every $m \in \mathcal{V}^L$.

We restate the optimality result of compositional languages under random noise from Kuciński et al. (2021) below.

Proposition 6.6 (Optimality under random noise (Kuciński et al., 2021)). Suppose that $|\mathcal{X}| = |\mathcal{V}| \geq 2$ and $N = L \geq 2$. Let $0 < \epsilon < \frac{|\mathcal{V}|-1}{|\mathcal{V}|}$. A one-to-one language $\ell : \mathcal{X}^N \rightarrow \mathcal{V}^L$ attains the maximum expected payoff under random noise Q_ϵ from Definition 6.3 if and only if it is compositional.

We also take a similar approach to Ananthakrishnan et al. (2025) and analyze the learning dynamics of the communication game through the lens of online learning. Each step is carried out as follows and repeated T times:

- (i) The sender and receiver jointly choose a language $\ell : \mathcal{X}^N \rightarrow \mathcal{V}^L$ from the set of all possible one-to-one languages.
- (ii) An object x is drawn from the uniform distribution over $\mathcal{X}^N$.
- (iii) The sender encodes the object into a message as $m = \ell(x) \in \mathcal{V}^L$.
- (iv) The message m is modified by a corruption scheme as $m' \sim Q(\cdot | m)$.
- (v) The receiver decodes the corrupted message as $x' = \ell^{-1}(m')$.
- (vi) The sender and receiver receive payoff $\frac{1}{N} \sum_{n=1}^N \mathbb{1}[x'_n = x_n]$.

We analyze the learning dynamics of this game through the expected regret defined below.

Definition 6.7 (Expected regret). Let $\mathcal{L}$ be the set of one-to-one languages from $\mathcal{X}^N$ to $\mathcal{V}^L$. Let $w : \mathcal{L} \rightarrow [0, 1]$ be an expected payoff function over languages. At each round t , a learning rule chooses a distribution $\mathbb{P}_w^{(t)}$ over $\mathcal{L}$. The expected regret after T rounds is defined as

$$\text{Reg}_T(w) := T \max_{\ell \in \mathcal{L}} w(\ell) - \sum_{t=1}^T \sum_{\ell \in \mathcal{L}} \mathbb{P}_w^{(t)}(\ell) w(\ell).$$

7. Misinformation

Setup We first consider the environment where the intermediary is interested in the entire set of attributes, i.e., $\mathcal{S} = \{1, 2, \dots, N\}$.

Can misinformation help induce compositional languages? We show that compositional languages are optimal under misinformation in a similar manner to Proposition

6.6 (Kuciński et al., 2021) that proves the optimality of compositional languages under random noise.

Proposition 7.1 (Minimax optimality of compositional languages). Suppose that $|\mathcal{X}| = |\mathcal{V}| \geq 2$ and $N = L \geq 2$. Let $\mathcal{L}$ be the set of one-to-one languages from $\mathcal{X}^N$ to $\mathcal{V}^L$. Let $k \in \left(0, \frac{|\mathcal{V}|-1}{|\mathcal{V}|}L\right)$. Then

$$\max_{\ell \in \mathcal{L}} \min_{Q \in \mathcal{Q}_k} J(\ell, Q)$$

is attained if and only if ℓ is a compositional language. In particular, for any compositional and non-compositional languages $\ell^*, \tilde{\ell} \in \mathcal{L}$,

$$\min_{Q \in \mathcal{Q}_k} J(\ell^*, Q) > \min_{Q \in \mathcal{Q}_k} J(\tilde{\ell}, Q).$$

We provide proof for Proposition 7.1 in Appendix A.

Can misinformation exert stronger pressure than random noise? We analyze how misinformation can exert stronger pressure toward the emergence of compositional languages compared to random noise by studying the learning dynamics in an online learning framework.

Proposition 7.2. Suppose that $|\mathcal{X}| = |\mathcal{V}| \geq 2$ and $N = L \geq 2$. Let $k \in \left(0, \frac{|\mathcal{V}|-1}{|\mathcal{V}|}L\right)$ and $\epsilon = \frac{k}{L}$. Let $\mathcal{L}$ denote the set of one-to-one languages from $\mathcal{X}^N$ to $\mathcal{V}^L$. Denote the expected payoff of languages under misinformation and random noise as

$$u(\ell) := \min_{Q \in \mathcal{Q}_k} J(\ell, Q) \quad \text{and} \quad v(\ell) := J(\ell, Q_\epsilon).$$

Define the compositionality gap under $w \in \{u, v\}$ as

$$\Delta_w := \max_{\ell^* \in \mathcal{L}_{\text{comp}}} w(\ell^*) - \max_{\tilde{\ell} \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} w(\tilde{\ell}),$$

where $\mathcal{L}_{\text{comp}} \subset \mathcal{L}$ is the set of compositional languages. Define the expected usage rate of compositional languages under $w \in \{u, v\}$ as

$$C_T(w) := \frac{1}{T} \sum_{t=1}^T \sum_{\ell \in \mathcal{L}_{\text{comp}}} \mathbb{P}_w^{(t)}(\ell).$$

Let Reg_T^* and Reg_T^ϵ be upper bounds on the learner's regrets under u and v :

$$\text{Reg}_T(u) \leq \text{Reg}_T^* \quad \text{and} \quad \text{Reg}_T(v) \leq \text{Reg}_T^\epsilon.$$

Then, misinformation yields a strictly tighter lower bound on compositional language usage compared to random noise whenever $\text{Reg}_T^* \leq \text{Reg}_T^\epsilon$ with $\text{Reg}_T^* > 0$,

$$C_T(u) \geq 1 - \frac{\text{Reg}_T^*}{\Delta_u T} \quad \text{and} \quad C_T(v) \geq 1 - \frac{\text{Reg}_T^\epsilon}{\Delta_v T},$$

as

$$\Delta_u - \Delta_v \geq \frac{3}{4N|\mathcal{X}|^N} \left(\frac{k}{N(|\mathcal{X}|-1)} \right)^N > 0.$$

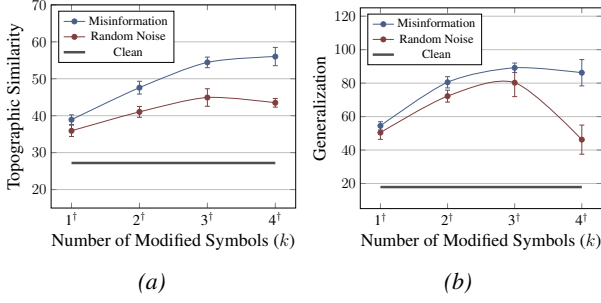


Figure 1. Language properties under different channel conditions. Statistically significant differences for misinformation relative to random noise are denoted by †.

We provide proof for Proposition 7.2 in Appendix A. Proposition 7.2 suggests that misinformation can more strongly pressure the sender and receiver to use compositional languages than random noise and that the difference between the two becomes more pronounced as the number of modified symbols, k , increases. We note that a language is non-compositional if there exists a single violating pair of mappings, and the worst-case bound expression for the gap $\Delta_u - \Delta_v$ decreases as the overall sizes of object and message spaces increase. As for the regret bounds Reg_T^* and Reg_T^ϵ , gap-agnostic worst-case regret bounds that do not depend on the payoff gap with respect to the best, i.e., compositional, languages, the regret bounds for misinformation and random noise can be identical. For example, a standard bound for Hedge algorithm is $\mathcal{O}(\sqrt{T \log |\mathcal{L}|})$ (Freund & Schapire, 1997). For gap-dependent regret bounds based on the difference in expected payoffs between compositional and non-compositional languages, misinformation can help attain a tighter regret upper bound expression than random noise. For example, in the case of the standard regret bound of UCB1 (Auer et al., 2002), misinformation yields a strictly smaller upper bound expression than random noise whenever $T \geq 2$. See Corollary A.6 in Appendix A for details.

Misinformation can promote compositionality We empirically analyze the effect of misinformation. In Figure 1a, we observe that misinformation tends to induce high scores of TopSim compared to random noise, suggesting that languages developed under risks of malign misrepresentation are more compositional. The differences get more pronounced as the number of modified symbols, k , is increased. We also observe a similar trend in generalization ability from Figure 1b. We hypothesize that the malign intermediary’s ability to more effectively exploit vulnerabilities of non-distributional expressions in a language forces the language to be more compositional.

8. Misinformation under new intermediaries

What if the intermediaries are not adaptive? Proposition 7.2 assumes that the malign intermediary would always find the optimal corruption scheme for a given language ℓ at every time step. However, as sender and receiver adapt their language to circumvent the intermediary’s corruption scheme, they may find an exploitative high-payoff language for that specific corruption scheme. Proposition 8.1 shows that a corruption scheme that achieves lower expected payoff than random noise for a non-compositional language can be exploited by another non-compositional language to achieve higher expected payoff than compositional languages. This implies that a non-compositional language can be preferred over a compositional language unless the corruption scheme adapts to the protocol change, suggesting that an insufficiently adaptive intermediary may fail to sustain a consistent pressure toward compositionality.

Proposition 8.1 (Existence of exploitative non-compositional language). *Suppose that $|\mathcal{X}| = |\mathcal{V}| \geq 2$ and $N = L \geq 3$. Let $K \in \{2, \dots, L-1\}$ and $K_{\min} = K \left(1 - \left(1 - \frac{K}{N}\right)^N\right)$. For every $k \in (K_{\min}, K)$, set $\epsilon = \frac{k}{L}$ so that $Q_\epsilon \in \mathcal{Q}_k$. Then there exist two non-compositional languages $\ell', \tilde{\ell}$, and a corruption scheme $Q' \in \mathcal{Q}_k$, such that*

$$\underbrace{J(\ell', Q') < J(\ell', Q_\epsilon)}_{\text{Lower payoff than } Q_\epsilon} \quad \text{and} \quad \underbrace{J(\tilde{\ell}, Q') > J(\ell^*, Q')}_{\text{Higher payoff for non-compo.}}$$

where ℓ^* is any compositional language.

We provide proof for Proposition 8.1 in Appendix A.

Introduction of new intermediaries can foster compositionality Iterated learning studies language transmission across generations and finds that transmission bottlenecks can drive the emergence of compositionality (Smith et al., 2003). In neural network simulations, it is often instantiated as periodically introducing a newly initialized receiver during training (Li & Bowling, 2019; Ren et al., 2020). Inspired by this, we introduce new generations of malign intermediaries and test whether this fosters more adaptive pressure toward compositionality. To test this, the intermediary’s parameters are reinitialized at six evenly spaced intervals during training. In Figure 2a, we observe that languages developed under the intermediary reset regime tend to be more compositional than those developed under misinformation without resets. This suggests that the introduction of new intermediaries can help exert more consistent pressure toward the emergence of compositionality. We also observe higher degrees of positional disentanglement under the reset regime in Figure 2c and stronger compositional generalization in Figure 2b. We do not observe significant differences in BosDis in Figure 2d, suggesting that symbol-wise structures are not as effective in dealing with corruptions.

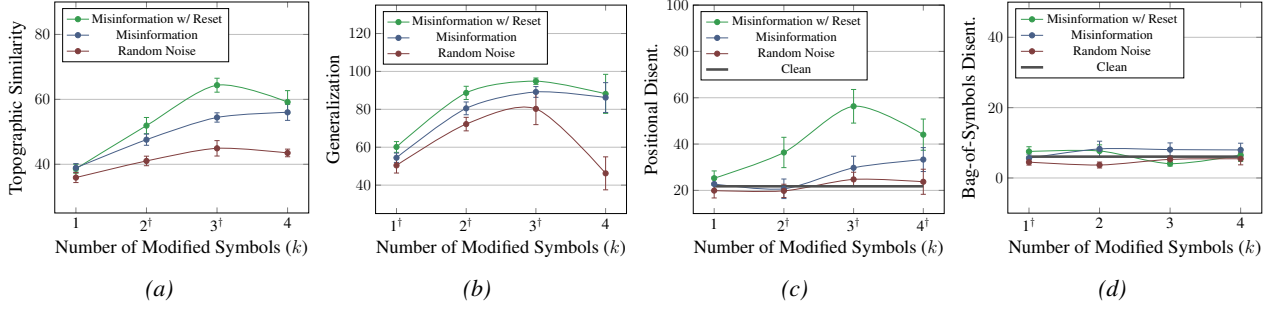


Figure 2. Language properties under misinformation with intermediary reset regime. Statistically significant differences for misinformation with reset relative to misinformation without reset are denoted by † .

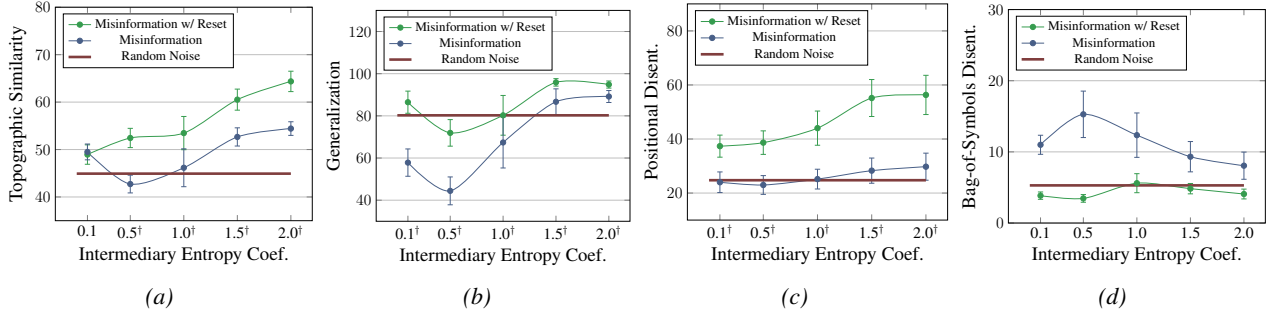


Figure 3. Properties of languages under varying intermediary entropy coefficients in misinformation with and without reset regimes. Statistically significant differences for misinformation with reset relative to misinformation without reset are denoted by † .

Non-adaptive intermediaries may hinder compositionality

To empirically examine the role of adaptability, we fix the modification budget to $k = 3$ and control adaptability of the intermediary by adjusting the entropy regularization coefficient in the intermediary’s loss function and observe how it affects the emergent languages. In Figure 3a, we observe that small coefficients yield lower scores of TopSim when intermediaries are not reset, and introduction of new intermediaries mitigates this effect. Similar trends are observed in PosDis in Figure 3c and compositional generalization in Figure 3b. These results align with the theoretical analysis in Proposition 8.1, suggesting that non-adaptive intermediaries may be suboptimal in inducing compositionality.

Languages induced under misinformation are easier to learn

We analyze how misinformation affects learnability of emergent languages by training newly initialized receivers to play the communication game with frozen senders trained under different channel conditions. In Figure 4, we find that languages induced under misinformation are easier to learn than languages induced under random noise, and that new receivers learn languages induced under the intermediary-reset setup the fastest. This suggests that misinformation can induce more structured and hence easier-to-learn languages, and that introduction of new intermediaries further promotes this effect.

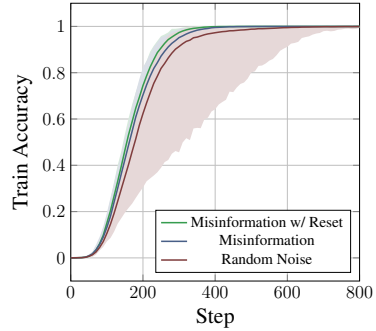


Figure 4. Learning curves of new receivers trained on languages induced under different conditions. Shaded regions denote minimum and maximum values across random seeds.

Can misinformation induce compositionality across different environments?

We experiment with different numbers of attributes, values per attribute, and message lengths to test whether the effects persist across different environments. For the attribute-value sizes, we change the number of attributes by one and control the number of values to keep the number of objects approximately constant, which yields $(N, |\mathcal{X}|)$ pairs of (4, 18) and (6, 7). For message length, we consider $L \in \{8, 9, 10, 11, 12\}$. We fix the number of modified symbols $k = 3$ where random noise exhibits the highest compositionality and generalization abilities. In Table 1 and Figure 5, we observe that misinformation tends

Metric	Condition	(4, 18)	(6, 7)
TopSim	Misinfo w/ Reset	55.2 $\pm 6.8^\dagger$	63.8 $\pm 4.1^\dagger$
	Misinformation	46.6 ± 2.7	55.4 $\pm 3.4^\ddagger$
	Random	46.6 ± 3.5	44.6 ± 3.4
	Clean	18.4 ± 2.4	32.3 ± 6.7
Generalization	Misinfo w/ Reset	86.6 ± 11.0	96.7 $\pm 5.5^\dagger$
	Misinformation	85.6 ± 7.8	87.1 $\pm 12.6^\ddagger$
	Random	89.6 ± 4.8	72.9 ± 11.3
	Clean	8.9 ± 7.8	24.0 ± 21.5
PosDis	Misinfo w/ Reset	60.3 $\pm 15.7^\dagger$	47.7 $\pm 8.7^\dagger$
	Misinformation	33.6 ± 9.5	26.7 $\pm 7.0^\ddagger$
	Random	33.4 ± 5.9	19.2 ± 5.6
	Clean	15.4 ± 5.0	20.1 ± 11.7

Table 1. Properties of languages induced from varying attribute-value configurations. **Bold** and underlined numbers indicate the highest and second-highest values in each column. Numbers to the right of $\pm$ denote standard deviations. Statistically significant differences relative to misinformation without reset and random noise are denoted by † and ‡ , respectively.

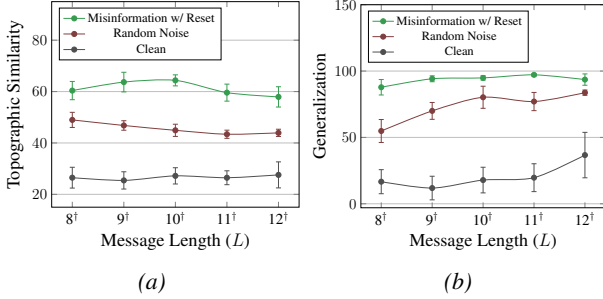


Figure 5. Emergent language properties under varying message lengths. Statistically significant differences for misinformation with reset relative to random noise are denoted by † .

to induce high degrees of compositionality across different attribute-value configurations and message lengths (see Table 2 and Figure 12 in Appendix D for full results). We observe similar trends under varying intercept probability and message-only intermediary conditions (see Figures 18 and 19 in Appendix D). We also experiment with image inputs in Appendix E and observe similar results.

9. Partial misinformation

Setup We explore partial misinformation environment where the malign intermediary is only rewarded for inducing incorrect predictions on a subset of attributes indexed by $\mathcal{S} \subseteq \{1, 2, \dots, N\}$. The intermediary is indifferent to the receiver’s predictions on attributes that lie outside $\mathcal{S}$. The sender and receiver are still rewarded for correct predictions on all attributes. This environment represents a stronger pressure toward the attributes of interest, under the same modification budget, but at the expense of ignoring other attributes. We compare partial misinformation with the same

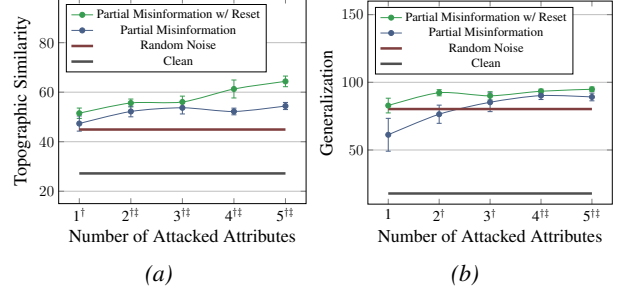


Figure 6. Properties of emergent languages by number of attacked attributes under partial misinformation. Statistically significant differences for misinformation with reset and without reset relative to random noise are denoted by † and ‡ , respectively.

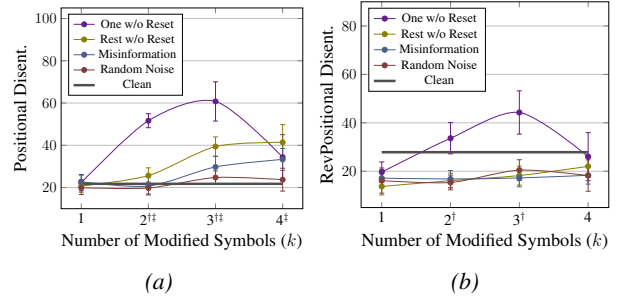


Figure 7. Properties of languages induced under the shifting interest setup without intermediary resets. Statistically significant differences relative to misinformation in the one-attribute and rest-attribute setups are denoted by † and ‡ , respectively.

random noise environment as in the preceding experiments.

Can partial misinformation foster compositionality?

We first explore whether partial misinformation induces compositionality. In this setup, we vary the size of the target attribute set, $\mathcal{S}$, from 1 to 5 and test whether partial misinformation can also promote compositionality. We fix the number of modified symbols $k = 3$. In the intermediary reset regime, the intermediaries’ parameters are reinitialized at six evenly spaced intervals during training. In Figure 6a, we observe that partial misinformation can also induce compositionality. Similar patterns are observed in generalization ability in Figure 6b. These results suggest that attacking all attributes is not necessary for the emergence of compositionality, and that even an intermediary that is interested in part of the information can exert meaningful pressure. See Figure 13 for full results in Appendix D.

How do shifting interests affect compositionality?

We also consider a shifting interest setup, where the set of targeted attribute indices $\mathcal{S}$ changes during learning. Specifically, in the *one-attribute* setting, the target attribute index set $\mathcal{S}$ is sequentially changed from $\{1\}$ to $\{5\}$ at five evenly spaced intervals during training. In the *rest-attribute* setting, $\mathcal{S}$ is sequentially changed from $[N] \setminus \{1\}$ to $[N] \setminus \{5\}$,

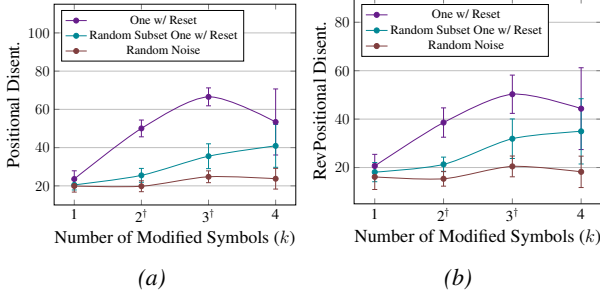


Figure 8. Properties of languages induced under per-step random interests setup. Statistically significant differences for one w/ reset relative to random subset one w/ reset are denoted by [†].

where $[N] = \{1, \dots, N\}$, at the same intervals as in the *one-attribute* setting. We consider both with and without intermediary reset regimes as we expect the shifting interests could have similar effects to those of parameter reinitialization. In Figures 7a and 7b, we observe that intermediaries with shifting interests induce high degrees of position-wise structures. We speculate that this is because intermediaries would be more likely to attack positions where the target attributes reside, thus creating an incentive for the sender and receiver to isolate other non-target attributes in separate positions. Appendices F.3 and F.4 qualitatively analyze how partial misinformation can induce positional separation of attributes. See Figures 14 and 15 in Appendix D for full results.

Prolonged exposure is a critical factor To better understand the conditions under which partial misinformation can promote compositionality, we design an environment where, at each time step, the intermediary takes the target attribute indices $\mathcal{S}$ along with the message m and the object x as input. We take the *one-attribute* setting. The target attribute index set $\mathcal{S}$ is randomly sampled from $\{\{1\}, \dots, \{5\}\}$ at each time step during training. This design removes prolonged exposure to attacks on a set of attributes and instead creates a temporally more distributed pressure across all attributes. In Figures 8a and 8b, we observe that this form of partial misinformation induces lower degrees of compositionality than the prolonged pressure environment, suggesting that prolonged exposure to a specific set of attributes is a crucial factor for partial misinformation to effectively promote compositionality. See Figure 16 for full results in Appendix D.

Biased random noise fails to facilitate compositionality We hypothesize that malign intermediaries interested in only a subset of attributes may modify symbols in a positionally biased manner. We test whether positionally biased random noise induces similar effects to those of partial misinformation by designing two types of biased random noise. In the *static bias* setting, the symbols at a fixed set of

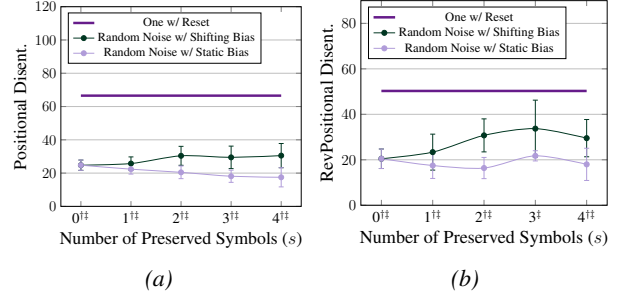


Figure 9. Language properties under varying numbers of preserved symbols. Statistically significant differences for one w/ reset relative to shifting and static biases are denoted by [†] and [‡], respectively.

$L - s$ positions in each message are subject to random noise modification, while the remaining s positions are always preserved. In the *shifting bias* setting, the set of s preserved positions is randomly resampled at five evenly spaced intervals during training. We set the number of modified symbols $k = 3$ and compare the two types of biased random noise with the *one-attribute* partial misinformation setting. In Figures 9a and 9b, we observe that neither static nor shifting bias induces degrees of compositionality comparable to those induced by malign modification. The results indicate that the effects of partial misinformation are not simply due to positional biases in the modifications, but to a directed pressure on specific attributes. We hypothesize that this is because the sender and receiver are simply incentivized to store information in preserved positions regardless of the attribute. In contrast, under partial misinformation, positional separation of target attributes may create safe positions for non-target attributes. See Figure 17 in Appendix D for full results.

10. Conclusion

This work draws a connection between misinformation and the emergence of compositionality in languages. We find that directed attempts at misrepresentation can lead the agents to develop more structured languages than undirected random corruptions. We provide a theoretical account of how misinformation can promote compositionality and empirically support the hypothesis that this effect arises from the malign intermediary’s ability to exploit non-compositional vulnerabilities in a language. We identify the intermediary’s adaptability as a crucial factor in sustaining consistent pressure toward compositionality and find that intermediaries that target a subset of attributes can also promote compositionality. These findings suggest that the commonly observed acts of misinformation in social interactions may serve as an important environmental pressure in the emergence of compositional languages.

Impact statement

This work explores the effects of misinformation on emergent communication systems. Understanding how misinformation influences communication protocols can provide insights into designing robust and compositional communication systems in multi-agent environments and in broader machine learning contexts. This section provides a discussion of potential applications of the findings and suggestions for future research directions that can be built upon this work.

Implications for broader machine learning Similar to prior works (Zheng et al., 2024; Ren et al., 2023; Noukhovitch et al., 2023) that leverage the insights from emergent communication studies in general machine learning research, applications of this work’s findings to solving practical problems could be explored. Future research may examine the effects of misinformation in communication among large language model (LLM) agents (Davidson et al., 2025) or interactions between LLMs and reward models in reinforcement learning from human feedback (Ouyang et al., 2022) to investigate whether misinformation encourages models to use more structured expressions. Zheng et al. (2024) framed the similarity computation between text and image embeddings in vision-language models (VLMs) (Radford et al., 2021) as a form of communication and applied iterated learning on the text encoder side to improve compositionality of VLMs. Adversarial modifications of these embeddings could be similarly explored. This work’s findings suggest that adversarial interception of messages in multi-agent reinforcement learning environments may help newly introduced partners learn existing communication protocols more quickly by promoting the emergence of compositional, hence easy-to-learn, protocols. They also indicate that the adaptability of the modification scheme is crucial for sustaining this pressure. The partial misinformation results further suggest that such interventions need not target all information contained in a message.

Misinformation within the complexity of real-world languages and environments Compositionality remains a nuanced concept that is difficult to precisely define or measure (Elmoznino et al., 2025; Korbak et al., 2020). The compositionality measures used in this work emphasize linear structures and may not fully reflect hierarchical or nested structures or structures that express negations of a concept. This work also employs synthetic datasets with a simple reconstruction task to explore how compositional structures emerge under misinformation. While this allows a controlled analysis of the effects of misinformation, there exist discrepancies between the synthetic and real-world environments. Also, the neural networks employed in this work may not fully reflect human cognition and its complex

learning dynamics. The types of misinformation that promote more complex compositional structures in real-world environments could differ from the ones that this work considers. Future work may explore more diverse types of misinformation and their relationship to the emergence of more nuanced compositional characteristics in languages.

Flexible theoretical framework The theoretical analysis in this work assumes one-to-one languages and dimension alignments between objects and messages. These assumptions allow an exact analytical comparison, in that changes in a message’s symbols correspond cleanly to changes in the decoded meaning. While similar assumptions are employed in more analytical explorations of linguistic properties (Kuciński et al., 2021; Ren et al., 2020), they nonetheless remain simplifications of real-world communication protocols. The online learning setting in the theoretical analysis also exhibits deviations from real-world scenarios. In the online learning setting, the differences between languages are not as strictly constrained, as the distribution over languages can choose a completely different language from the language chosen in the previous step. This would not be plausible in a neural network or a human, as it would require a large shift in its internal structure. Flexible theoretical frameworks that can accommodate more complex language structures and learning dynamics could provide a more comprehensive understanding of the effects of misinformation on the emergence of compositionality.

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A. Proofs for theoretical results

Lemma A.1 (Equivalence of compositionality and global distance preservation). *Suppose $|\mathcal{X}| = |\mathcal{V}|$ and $N = L$. A one-to-one language $\ell : \mathcal{X}^N \rightarrow \mathcal{V}^L$ is compositional, if and only if for any $x_1, x_2 \in \mathcal{X}^N$,*

$$\rho(x_1, x_2) = c \iff \rho(\ell(x_1), \ell(x_2)) = c,$$

where ρ is the Hamming distance.

Proof of Lemma A.1. (I) Distance-1 isometry holds in both directions

Fix a compositional one-to-one language ℓ . Consider distance-1 neighbors of arbitrary x and $\ell(x)$:

$$\mathcal{N}_{\mathcal{X}^N} = \{x' \in \mathcal{X}^N : \rho(x, x') = 1\}, \quad \mathcal{N}_{\mathcal{V}^L} = \{m' \in \mathcal{V}^L : \rho(\ell(x), m') = 1\}.$$

By the definition of compositionality,

$$\ell(x') \in \mathcal{N}_{\mathcal{V}^L} \quad \text{for any } x' \in \mathcal{N}_{\mathcal{X}^N},$$

in other words,

$$\ell(\mathcal{N}_{\mathcal{X}^N}) \subseteq \mathcal{N}_{\mathcal{V}^L}.$$

Since $|\mathcal{X}| = |\mathcal{V}|$ and $N = L$, they have the same number of neighbors:

$$|\mathcal{N}_{\mathcal{X}^N}| = N(|\mathcal{X}| - 1) = L(|\mathcal{V}| - 1) = |\mathcal{N}_{\mathcal{V}^L}|.$$

$\ell(\mathcal{N}_{\mathcal{X}^N}) \subseteq \mathcal{N}_{\mathcal{V}^L}$ is between two finite sets of the same cardinality. As ℓ is a bijection, every $x' \in \mathcal{N}_{\mathcal{X}^N}$ is mapped to a unique $\ell(x') \in \mathcal{N}_{\mathcal{V}^L}$, implying that

$$\ell(\mathcal{N}_{\mathcal{X}^N}) = \mathcal{N}_{\mathcal{V}^L}.$$

Therefore, for every $x_1, x_2 \in \mathcal{X}^N$,

$$\rho(x_1, x_2) = 1 \iff \rho(\ell(x_1), \ell(x_2)) = 1.$$

(II) Distance-1 isometry implies global isometry

For any $x_1, x_2 \in \mathcal{X}^N$ with $\rho(x_1, x_2) = c$, there exists a sequence of objects $x^{(i)} \in \mathcal{X}^N$,

$$x_1 = x^{(0)}, x^{(1)}, x^{(2)}, \dots, x^{(c-1)}, x^{(c)} = x_2, \quad \text{such that } \rho(x^{(i-1)}, x^{(i)}) = 1 \text{ for } i = 1, \dots, c.$$

Applying ℓ to each object $x^{(i)}$, we have

$$\ell(x_1) = \ell(x^{(0)}), \ell(x^{(1)}), \ell(x^{(2)}), \dots, \ell(x^{(c-1)}), \ell(x^{(c)}) = \ell(x_2).$$

As ℓ is compositional, this repeats single coordinate change c times, hence

$$\rho(\ell(x_1), \ell(x_2)) \leq c$$

We can construct a similar sequence on the message side, by denoting $d := \rho(\ell(x_1), \ell(x_2))$,

$$\ell(x_1) = m^{(0)}, m^{(1)}, m^{(2)}, \dots, m^{(d-1)}, m^{(d)} = \ell(x_2), \quad \text{such that } \rho(m^{(i-1)}, m^{(i)}) = 1 \text{ for } i = 1, \dots, d.$$

Applying ℓ^{-1} to each message $m^{(i)}$, we have

$$x_1 = \ell^{-1}(m^{(0)}), \ell^{-1}(m^{(1)}), \ell^{-1}(m^{(2)}), \dots, \ell^{-1}(m^{(d-1)}), \ell^{-1}(m^{(d)}) = x_2.$$

Results from (I) imply $\rho(\ell^{-1}(m^{(i-1)}), \ell^{-1}(m^{(i)})) = 1$. By triangle inequality, we have

$$\rho(x_1, x_2) \leq d.$$

Combining the two inequalities yields

$$d = \rho(\ell(x_1), \ell(x_2)) \leq c = \rho(x_1, x_2) \leq d.$$

Therefore, for any $x_1, x_2 \in \mathcal{X}^N$,

$$\rho(x_1, x_2) = c \iff \rho(\ell(x_1), \ell(x_2)) = c.$$

Conversely, if global isometry holds, i.e., for any $x_1, x_2 \in \mathcal{X}^N$,

$$\rho(\ell(x_1), \ell(x_2)) = c \iff \rho(x_1, x_2) = c,$$

then, in particular, compositionality ($c = 1$) is satisfied:

$$\rho(x_1, x_2) = 1 \iff \rho(\ell(x_1), \ell(x_2)) = 1.$$

□

Lemma A.2 (Existence of a scrambled association). *Suppose that $|\mathcal{X}| = |\mathcal{V}| \geq 2$ and $N = L \geq 2$. Let ℓ be a non-compositional one-to-one language, $\ell : \mathcal{X}^N \rightarrow \mathcal{V}^L$. Then, there exist $x_0, x_1, x_2 \in \mathcal{X}^N$ and their corresponding messages $m_0 = \ell(x_0), m_1 = \ell(x_1), m_2 = \ell(x_2)$ such that*

$$\rho(x_0, x_1) = \rho(m_0, m_2) = c > 0, \quad \text{and} \quad \rho(x_0, x_2) \neq c, \rho(m_0, m_1) \neq c.$$

Proof of Lemma A.2. Fix a non-compositional language ℓ . Since ℓ is not compositional, by Lemma A.1, there exist $x_0, x_1 \in \mathcal{X}^N$ and their corresponding messages $m_0 = \ell(x_0)$ and $m_1 = \ell(x_1)$ such that,

$$\rho(x_0, x_1) = c > 0 \quad \text{and} \quad \rho(m_0, m_1) \neq c.$$

Consider the distance- c neighbors of x_0 and m_0 :

$$\mathcal{N}_{\mathcal{X}^N}(x_0, c) = \{x' \in \mathcal{X}^N : \rho(x_0, x') = c\}, \quad \mathcal{N}_{\mathcal{V}^L}(m_0, c) = \{m' \in \mathcal{V}^L : \rho(m_0, m') = c\}.$$

Since $|\mathcal{X}| = |\mathcal{V}|$ and $N = L$, the sizes of the two neighbor sets are equal:

$$|\mathcal{N}_{\mathcal{X}^N}(x_0, c)| = \binom{N}{c} (|\mathcal{X}| - 1)^c = \binom{L}{c} (|\mathcal{V}| - 1)^c = |\mathcal{N}_{\mathcal{V}^L}(m_0, c)|.$$

As $m_1 \notin \mathcal{N}_{\mathcal{V}^L}(m_0, c)$,

$$\ell(\mathcal{N}_{\mathcal{X}^N}(x_0, c)) \neq \mathcal{N}_{\mathcal{V}^L}(m_0, c).$$

Note that ℓ is a bijection, hence

$$|\ell(\mathcal{N}_{\mathcal{X}^N}(x_0, c))| = |\mathcal{N}_{\mathcal{X}^N}(x_0, c)| = |\mathcal{N}_{\mathcal{V}^L}(m_0, c)|.$$

So there exists $m_2 \in \mathcal{N}_{\mathcal{V}^L}(m_0, c) \setminus \ell(\mathcal{N}_{\mathcal{X}^N}(x_0, c))$ such that

$$x_2 = \ell^{-1}(m_2) \notin \mathcal{N}_{\mathcal{X}^N}(x_0, c).$$

Therefore, we have

$$\rho(x_0, x_1) = \rho(m_0, m_2) = c > 0, \quad \text{and} \quad \rho(x_0, x_2) \neq c, \rho(m_0, m_1) \neq c.$$

□

Lemma A.3 (Bounding property of the maximum scrambled distance). *Suppose that $|\mathcal{X}| = |\mathcal{V}| \geq 2$ and $N = L \geq 2$. Let ℓ be a non-compositional one-to-one language, $\ell : \mathcal{X}^N \rightarrow \mathcal{V}^L$. Consider $x_0, x_1, x_2 \in \mathcal{X}^N$ and their corresponding messages $m_0 = \ell(x_0), m_1 = \ell(x_1), m_2 = \ell(x_2)$ such that*

$$\rho(x_0, x_1) = \rho(m_0, m_2) = c > 0 \quad \text{and} \quad \rho(x_0, x_2) \neq c, \rho(m_0, m_1) \neq c.$$

Choose one for which c is maximal. Then

$$\rho(x_0, x_2) < c \quad \text{and} \quad \rho(m_0, m_1) < c.$$

Proof of Lemma A.3. We use the same argument as in Lemma A.2. Let $a := \rho(x_0, x_2)$ and $b := \rho(m_0, m_1)$. Suppose for contradiction that $a > c$. Consider distance- a neighbors of x_0 and m_0 :

$$\mathcal{N}_{\mathcal{X}^N}(x_0, a) = \{x' \in \mathcal{X}^N : \rho(x_0, x') = a\}, \quad \mathcal{N}_{\mathcal{V}^L}(m_0, a) = \{m' \in \mathcal{V}^L : \rho(m_0, m') = a\}.$$

Since ℓ is a one-to-one language, we have

$$|\ell(\mathcal{N}_{\mathcal{X}^N}(x_0, a))| = |\mathcal{N}_{\mathcal{X}^N}(x_0, a)|.$$

Because $|\mathcal{X}| = |\mathcal{V}|$ and $N = L$, we have

$$|\mathcal{N}_{\mathcal{X}^N}(x_0, a)| = |\mathcal{N}_{\mathcal{V}^L}(m_0, a)|.$$

Hence, we get

$$|\ell(\mathcal{N}_{\mathcal{X}^N}(x_0, a))| = |\mathcal{N}_{\mathcal{V}^L}(m_0, a)|.$$

$a \neq c$ implies that $\ell(x_2) \notin \mathcal{N}_{\mathcal{V}^L}(m_0, a)$, but since $x_2 \in \mathcal{N}_{\mathcal{X}^N}(x_0, a)$ and $|\ell(\mathcal{N}_{\mathcal{X}^N}(x_0, a))| = |\mathcal{N}_{\mathcal{V}^L}(m_0, a)|$, we have

$$\mathcal{N}_{\mathcal{V}^L}(m_0, a) \setminus \ell(\mathcal{N}_{\mathcal{X}^N}(x_0, a)) \neq \emptyset.$$

So, there exists m_3 such that

$$m_3 \in \mathcal{N}_{\mathcal{V}^L}(m_0, a) \setminus \ell(\mathcal{N}_{\mathcal{X}^N}(x_0, a)).$$

Let $x_3 := \ell^{-1}(m_3)$. Note that $m_3 \notin \ell(\mathcal{N}_{\mathcal{X}^N}(x_0, a))$, so $x_3 \notin \mathcal{N}_{\mathcal{X}^N}(x_0, a)$. This implies

$$\begin{aligned} \rho(m_0, m_3) &= a > c \quad \text{and} \quad \rho(x_0, x_3) \neq a, \\ \rho(x_0, x_2) &= a > c \quad \text{and} \quad \rho(m_0, m_2) = c \neq a. \end{aligned}$$

Therefore, x_0, x_2, x_3 and their corresponding messages m_0, m_2, m_3 satisfy the conditions in the lemma with a larger value of c , which contradicts the maximality of c . Hence, $\rho(x_0, x_2) < c$ must hold. By the symmetry of $\mathcal{X}^N$ and $\mathcal{V}^L$ and ℓ being a one-to-one language, roles of object and message spaces can be interchanged, so the same argument applies to $\rho(m_0, m_1) < c$. $\square$

Lemma A.4 (Existence of superior corruption scheme). *Suppose that $|\mathcal{X}| = |\mathcal{V}| \geq 2$ and $N = L \geq 2$. Consider the random noise corruption scheme Q_ϵ defined as*

$$Q_\epsilon(m' | m) = (1 - \epsilon)^{L - \rho(m', m)} \epsilon^{\rho(m', m)} \left(\frac{1}{|\mathcal{V}| - 1} \right)^{\rho(m', m)},$$

where $0 < \epsilon = \frac{k}{L} < \frac{|\mathcal{V}| - 1}{|\mathcal{V}|}$. Define the set of all allowed corruption schemes as

$$\mathcal{Q}_k = \{Q : \mathbb{E}_{m' \sim Q(\cdot | m)} \rho(m', m) = k, \forall m \in \mathcal{V}^L\}.$$

Then, for any non-compositional one-to-one language $\tilde{\ell}$, there exists a corruption scheme $Q \in \mathcal{Q}_k$ such that the difference of the expected payoff of $\tilde{\ell}$ under Q and Q_ϵ is lower bounded by

$$J(\tilde{\ell}, Q_\epsilon) - J(\tilde{\ell}, Q) \geq \frac{3}{4N|\mathcal{X}|^N} \left(\frac{k}{N(|\mathcal{X}| - 1)} \right)^N,$$

where J is the expected payoff function defined in Definition 6.4.

Proof of Lemma A.4. (I) Selecting scrambled object-message associations

Fix a non-compositional one-to-one language $\tilde{\ell}$. By Lemma A.2, there exists at least one triple $x_0, x_1, x_2 \in \mathcal{X}^N$ with corresponding messages $m_i = \tilde{\ell}(x_i)$ such that

$$\rho(x_0, x_1) = \rho(m_0, m_2) = c > 0 \quad \text{and} \quad \rho(x_0, x_2) \neq c, \rho(m_0, m_1) \neq c.$$

Among all such triples, choose one that maximizes c , and let c^* denote this maximum value. By Lemma A.3, we then have

$$\rho(x_0, x_2) < \rho(x_0, x_1) = c^*, \quad \rho(m_0, m_1) < \rho(m_0, m_2) = c^*.$$

(II) Constructing a candidate corruption scheme Q

We modify Q_ϵ to construct a corruption scheme Q given m_0 as

$$Q(m' | m_0) = \begin{cases} Q_\epsilon(m_1 | m_0) + \delta, & \text{if } m' = m_1, \\ Q_\epsilon(m_2 | m_0) - \delta \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)}, & \text{if } m' = m_2, \\ Q_\epsilon(m_0 | m_0) - \delta \left(1 - \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)}\right), & \text{if } m' = m_0, \\ Q_\epsilon(m' | m_0), & m' \notin \{m_0, m_1, m_2\}, \end{cases}$$

where $\delta = \frac{1}{2}Q_\epsilon(m_2 | m_0)$. For $m \neq m_0$, we keep $Q(\cdot | m) = Q_\epsilon(\cdot | m)$ as is. Note that Q does not change the expected per-message modification from Q_ϵ :

$$\begin{aligned} \mathbb{E}_{m' \sim Q(\cdot | m_0)} \rho(m', m_0) - \mathbb{E}_{m' \sim Q_\epsilon(\cdot | m_0)} \rho(m', m_0) &= \delta \rho(m_0, m_1) - \delta \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)} \rho(m_0, m_2) - \delta \left(1 - \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)}\right) \cdot 0 \\ &= \delta \rho(m_0, m_1) - \delta \rho(m_0, m_1) \\ &= 0. \end{aligned}$$

The probabilities also sum to 1:

$$\sum_{m' \in \mathcal{V}^L} (Q(m' | m_0) - Q_\epsilon(m' | m_0)) = \delta - \delta \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)} - \delta \left(1 - \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)}\right) = \delta - \delta = 0.$$

We verify that the probabilities lie within $[0, 1]$. First, consider m_2 ,

$$\begin{aligned} Q(m_2 | m_0) &= Q_\epsilon(m_2 | m_0) - \frac{1}{2}Q_\epsilon(m_2 | m_0) \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)} \\ &= Q_\epsilon(m_2 | m_0) \left(1 - \frac{\rho(m_0, m_1)}{2\rho(m_0, m_2)}\right) \\ &> 0 \end{aligned}$$

as $\rho(m_0, m_1) < \rho(m_0, m_2)$ and $0 < \frac{\rho(m_0, m_1)}{2\rho(m_0, m_2)} < \frac{1}{2}$.

For m_0 , we have

$$Q(m_0 | m_0) = Q_\epsilon(m_0 | m_0) - \frac{1}{2}Q_\epsilon(m_2 | m_0) \left(1 - \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)}\right).$$

Since $0 < 1 - \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)} < 1$, we get

$$\frac{1}{2}Q_\epsilon(m_2 | m_0) \left(1 - \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)}\right) < \frac{1}{2}Q_\epsilon(m_2 | m_0).$$

Since m_2 lies at distance c from m_0 , $\rho(m_0, m_2) = c > 0$ and $\epsilon < \frac{|\mathcal{V}|-1}{|\mathcal{V}|}$, we have $Q_\epsilon(m_2 | m_0) < Q_\epsilon(m_0 | m_0)$ by monotonicity of $Q_\epsilon(m' | m_0)$ in $\rho(m', m_0)$. Therefore, $Q(m_0 | m_0) > 0$.

Note that since $Q(m_2 | m_0) > 0$ and $Q(m_0 | m_0) > 0$ and all of the probabilities sum to 1, each modified probability is at most 1, i.e., $Q(m_1 | m_0) \leq 1$.

Hence $Q \in \mathcal{Q}_k$.

(III) Verifying that Q is stronger than random noise

The expected per-attribute accuracy payoff of $\tilde{\ell}$ under Q for object x_0 is written as:

$$\begin{aligned}
 & \mathbb{E}_{m' \sim Q(\cdot | m_0)} \left[\frac{1}{N} \sum_{n=1}^N \mathbb{I} [\tilde{\ell}^{-1}(m')^{(n)} = x_0^{(n)}] \right] - \mathbb{E}_{m' \sim Q_\epsilon(\cdot | m_0)} \left[\frac{1}{N} \sum_{n=1}^N \mathbb{I} [\tilde{\ell}^{-1}(m')^{(n)} = x_0^{(n)}] \right] \\
 &= \frac{1}{N} \sum_{n=1}^N \mathbb{I} [x_1^{(n)} = x_0^{(n)}] \{Q(m_1 | m_0) - Q_\epsilon(m_1 | m_0)\} \\
 &\quad + \frac{1}{N} \sum_{n=1}^N \mathbb{I} [x_2^{(n)} = x_0^{(n)}] \{Q(m_2 | m_0) - Q_\epsilon(m_2 | m_0)\} \\
 &\quad + \frac{1}{N} \sum_{n=1}^N \mathbb{I} [x_0^{(n)} = x_0^{(n)}] \{Q(m_0 | m_0) - Q_\epsilon(m_0 | m_0)\} \\
 &= (1 - \frac{1}{N} \rho(x_0, x_1)) \{Q(m_1 | m_0) - Q_\epsilon(m_1 | m_0)\} \\
 &\quad + (1 - \frac{1}{N} \rho(x_0, x_2)) \{Q(m_2 | m_0) - Q_\epsilon(m_2 | m_0)\} \\
 &\quad + (1 - \frac{1}{N} \rho(x_0, x_0)) \{Q(m_0 | m_0) - Q_\epsilon(m_0 | m_0)\} \\
 &= \left(1 - \frac{1}{N} \rho(x_0, x_1)\right) \delta - \left(1 - \frac{1}{N} \rho(x_0, x_2)\right) \delta \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)} - \left(1 - \frac{1}{N} \rho(x_0, x_0)\right) \delta \left(1 - \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)}\right) \\
 &= \underbrace{\delta - \delta \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)} - \delta \left(1 - \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)}\right)}_{=0} - \frac{1}{N} \left\{ \rho(x_0, x_1) \delta - \rho(x_0, x_2) \delta \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)} \right\} \\
 &= -\frac{1}{N} \delta \left\{ \rho(x_0, x_1) - \rho(x_0, x_2) \underbrace{\frac{\rho(m_0, m_1)}{\rho(m_0, m_2)}}_{<1} \right\} \\
 &< 0.
 \end{aligned}$$

Since the expected per-attribute accuracy for x_0 is strictly lower under Q than Q_ϵ , the per-attribute accuracy payoff over $\text{Unif}(\mathcal{X}^N)$ is also strictly lower under Q than Q_ϵ , as they are identical for all $m \neq m_0$:

$$\mathbb{E}_{x \sim \text{Unif}(\mathcal{X}^N)} \mathbb{E}_{m' \sim Q(\cdot | m)} \frac{1}{N} \sum_{n=1}^N \mathbb{I}[x'_n = x_n] < \mathbb{E}_{x \sim \text{Unif}(\mathcal{X}^N)} \mathbb{E}_{m' \sim Q_\epsilon(\cdot | m)} \frac{1}{N} \sum_{n=1}^N \mathbb{I}[x'_n = x_n].$$

Note that Q only differs from Q_ϵ for m_0 . $\tilde{\ell}$ is a one-to-one language, so only x_0 is mapped to m_0 . Since the object distribution is uniform over $\mathcal{X}^N$, the probability of drawing x_0 is $\frac{1}{|\mathcal{X}|^N}$. Hence,

$$J(\tilde{\ell}, Q_\epsilon) - J(\tilde{\ell}, Q) = \frac{\delta}{|\mathcal{X}|^N N} \left\{ \rho(x_0, x_1) - \rho(x_0, x_2) \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)} \right\}.$$

(IV) Deriving a lower bound for $\rho(x_0, x_1) - \rho(x_0, x_2) \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)}$

Since

$$1 \leq \rho(x_2, x_0) \leq c^* - 1 \quad \text{and} \quad 1 \leq \rho(m_1, m_0) \leq c^* - 1,$$

we have

$$\rho(x_2, x_0) \rho(m_1, m_0) \leq (c^* - 1)^2.$$

Hence,

$$\begin{aligned}
 \rho(x_1, x_0) - \rho(x_2, x_0) \frac{\rho(m_1, m_0)}{\rho(m_2, m_0)} &= c^* - \frac{\rho(x_2, x_0)\rho(m_1, m_0)}{c^*} \\
 &\geq c^* - \frac{(c^* - 1)^2}{c^*} \\
 &= c^* - \frac{(c^*)^2 - 2c^* + 1}{c^*} \\
 &= c^* - c^* + 2 - \frac{1}{c^*} \\
 &= 2 - \frac{1}{c^*} \\
 &\geq 2 - \frac{1}{2} \\
 &= \frac{3}{2}.
 \end{aligned}$$

(V) *Deriving a lower bound for δ*

Note that

$$\epsilon < \frac{|\mathcal{V}| - 1}{|\mathcal{V}|} \Leftrightarrow |\mathcal{V}|\epsilon < |\mathcal{V}| - 1 \Leftrightarrow \epsilon < (|\mathcal{V}| - 1)(1 - \epsilon) \Leftrightarrow \frac{\epsilon}{|\mathcal{V}| - 1} < 1 - \epsilon.$$

The lower bound for $Q_\epsilon(m_2 | m_0)$ is hence given by,

$$\begin{aligned}
 Q_\epsilon(m_2 | m_0) &= (1 - \epsilon)^{L - c^*} \left(\frac{\epsilon}{|\mathcal{V}| - 1} \right)^{c^*} \\
 &\geq \left(\frac{\epsilon}{|\mathcal{V}| - 1} \right)^{L - c^*} \left(\frac{\epsilon}{|\mathcal{V}| - 1} \right)^{c^*} \\
 &= \left(\frac{\epsilon}{|\mathcal{V}| - 1} \right)^L.
 \end{aligned}$$

Therefore, the lower bound for δ is

$$\delta = \frac{1}{2} Q_\epsilon(m_2 | m_0) \geq \frac{1}{2} \left(\frac{\epsilon}{|\mathcal{V}| - 1} \right)^L.$$

(VI) *Combining the lower bounds*

Combining the lower bounds and noting that $\epsilon = \frac{k}{N}$, we have

$$\begin{aligned}
 J(\tilde{\ell}, Q_\epsilon) - J(\tilde{\ell}, Q) &\geq \frac{\delta}{|\mathcal{X}|^N N} \left\{ \rho(x_0, x_1) - \rho(x_0, x_2) \frac{\rho(m_0, m_1)}{\rho(m_0, m_2)} \right\} \\
 &\geq \frac{1}{2} \left(\frac{\epsilon}{|\mathcal{V}| - 1} \right)^L \left(\frac{1}{|\mathcal{X}|^N N} \right) \left(\frac{3}{2} \right) \\
 &= \frac{3}{4N|\mathcal{X}|^N} \left(\frac{\epsilon}{|\mathcal{V}| - 1} \right)^L \\
 &= \frac{3}{4N|\mathcal{X}|^N} \left(\frac{k}{N(|\mathcal{V}| - 1)} \right)^L \\
 &= \frac{3}{4N|\mathcal{X}|^N} \left(\frac{k}{N(|\mathcal{X}| - 1)} \right)^N.
 \end{aligned}$$

□

Lemma A.5 (Payoff upper bound under random noise). *Suppose that $|\mathcal{X}| = |\mathcal{V}| \geq 2$ and $N = L \geq 2$. Let Q_ϵ be the random noise from Definition 6.3 with $\epsilon = \frac{k}{L}$, where $0 < k < \frac{|\mathcal{V}|-1}{|\mathcal{V}|}L$. Let $J(\ell, Q_\epsilon, x)$ denote the expected payoff of language ℓ under corruption scheme Q_ϵ for object $x \in \mathcal{X}^N$, defined as*

$$J(\ell, Q_\epsilon, x) = \mathbb{E}_{m' \sim Q_\epsilon(\cdot | \ell(x))} \left[\frac{1}{N} \sum_{n=1}^N \mathbb{1} \left[\ell^{-1}(m')^{(n)} = x^{(n)} \right] \right],$$

where $x^{(n)}$ is the n -th attribute of object x . Then, for any object $x_0 \in \mathcal{X}^N$ and one-to-one language ℓ ,

$$J(\ell, Q_\epsilon, x_0) \leq 1 - \frac{k}{N},$$

and the equality holds if and only if

$$\rho(x_0, x) = \rho(\ell(x_0), \ell(x)) \quad \text{for all } x \in \mathcal{X}^N.$$

Proof of Lemma A.5. (I) $Q_\epsilon(m | m_0)$ is decreasing in $\rho(m, m_0)$

Fix an object $x_0 \in \mathcal{X}^N$ and denote $m_0 = \ell(x_0)$. The probability that the message m_0 is modified to m under Q_ϵ is given by

$$Q_\epsilon(m | m_0) = (1 - \epsilon)^{L - \rho(m, m_0)} \epsilon^{\rho(m, m_0)} \left(\frac{1}{|\mathcal{V}| - 1} \right)^{\rho(m, m_0)}.$$

The modification probability ratio of two messages $m_1, m_2 \in \mathcal{V}^L$ with $\rho(m_0, m_1) = \rho(m_0, m_2) - 1$ is given by

$$\frac{Q_\epsilon(m_1 | m_0)}{Q_\epsilon(m_2 | m_0)} = \frac{(1 - \epsilon)^{L - \rho(m_0, m_1)} \epsilon^{\rho(m_0, m_1)} \left(\frac{1}{|\mathcal{V}| - 1} \right)^{\rho(m_0, m_1)}}{(1 - \epsilon)^{L - \rho(m_0, m_2)} \epsilon^{\rho(m_0, m_2)} \left(\frac{1}{|\mathcal{V}| - 1} \right)^{\rho(m_0, m_2)}} = \frac{(1 - \epsilon)(|\mathcal{V}| - 1)}{\epsilon}.$$

Note that the ratio is greater than 1, as

$$\epsilon < \frac{|\mathcal{V}| - 1}{|\mathcal{V}|} \iff |\mathcal{V}| \epsilon < |\mathcal{V}| - 1 \iff \epsilon + (|\mathcal{V}| - 1) \epsilon < |\mathcal{V}| - 1 \iff \epsilon < (1 - \epsilon)(|\mathcal{V}| - 1) \iff 1 < \frac{(1 - \epsilon)(|\mathcal{V}| - 1)}{\epsilon}.$$

Therefore, $Q_\epsilon(m | m_0)$ is decreasing in $\rho(m, m_0)$.

(II) Distance- c message and object sets have the same cardinality

The number of messages that are at distance c from m_0 is

$$|\{m \in \mathcal{V}^L : \rho(m, m_0) = c\}| = \binom{L}{c} (|\mathcal{V}| - 1)^c.$$

As ℓ is a one-to-one language, the number of objects that are at distance c from x_0 is also given by

$$|\{x \in \mathcal{X}^N : \rho(x_0, x) = c\}| = \binom{N}{c} (|\mathcal{X}| - 1)^c.$$

Since $|\mathcal{X}| = |\mathcal{V}|$ and $N = L$, the number of messages that are at distance c from m_0 is equal to the number of objects that are at distance c from x_0 :

$$|\{m \in \mathcal{V}^L : \rho(m, m_0) = c\}| = |\{x \in \mathcal{X}^N : \rho(x_0, x) = c\}|.$$

(III) Scrambled association implies existence of language with a strictly higher payoff

Suppose that there exist two messages $m_1, m_2 \in \mathcal{V}^L$ such that

$$\rho(m_0, m_1) < \rho(m_0, m_2) \quad \text{and} \quad \rho(x_0, \ell^{-1}(m_1)) > \rho(x_0, \ell^{-1}(m_2)).$$

Define a new one-to-one language ℓ' by swapping the messages assigned to $\ell^{-1}(m_1)$ and $\ell^{-1}(m_2)$:

$$\ell'(x) = \begin{cases} m_2, & \text{if } x = \ell^{-1}(m_1), \\ m_1, & \text{if } x = \ell^{-1}(m_2), \\ \ell(x), & \text{otherwise.} \end{cases}$$

Let $x_1 = \ell^{-1}(m_1)$ and $x_2 = \ell^{-1}(m_2)$. ℓ' strictly increases the payoff under corruption scheme Q_ϵ for object x_0 :

$$\begin{aligned} J(\ell', Q_\epsilon, x_0) - J(\ell, Q_\epsilon, x_0) &= \mathbb{E}_{m' \sim Q_\epsilon(\cdot | m_0)} \left[\frac{1}{N} \sum_{n=1}^N \mathbb{1} [\ell'^{-1}(m')^{(n)} = x_0^{(n)}] \right] \\ &\quad - \mathbb{E}_{m' \sim Q_\epsilon(\cdot | m_0)} \left[\frac{1}{N} \sum_{n=1}^N \mathbb{1} [\ell^{-1}(m')^{(n)} = x_0^{(n)}] \right] \\ &= Q_\epsilon(m_2 | m_0) \left\{ \frac{1}{N} \sum_{n=1}^N \mathbb{1} [x_1^{(n)} = x_0^{(n)}] - \frac{1}{N} \sum_{n=1}^N \mathbb{1} [x_2^{(n)} = x_0^{(n)}] \right\} \\ &\quad + Q_\epsilon(m_1 | m_0) \left\{ \frac{1}{N} \sum_{n=1}^N \mathbb{1} [x_2^{(n)} = x_0^{(n)}] - \frac{1}{N} \sum_{n=1}^N \mathbb{1} [x_1^{(n)} = x_0^{(n)}] \right\} \\ &= Q_\epsilon(m_2 | m_0) \frac{1}{N} \sum_{n=1}^N \mathbb{1} [x_1^{(n)} = x_0^{(n)}] - Q_\epsilon(m_1 | m_0) \frac{1}{N} \sum_{n=1}^N \mathbb{1} [x_1^{(n)} = x_0^{(n)}] \\ &\quad + Q_\epsilon(m_1 | m_0) \frac{1}{N} \sum_{n=1}^N \mathbb{1} [x_2^{(n)} = x_0^{(n)}] - Q_\epsilon(m_2 | m_0) \frac{1}{N} \sum_{n=1}^N \mathbb{1} [x_2^{(n)} = x_0^{(n)}] \\ &= \{Q_\epsilon(m_2 | m_0) - Q_\epsilon(m_1 | m_0)\} \frac{1}{N} \sum_{n=1}^N \mathbb{1} [x_1^{(n)} = x_0^{(n)}] \\ &\quad + \{Q_\epsilon(m_1 | m_0) - Q_\epsilon(m_2 | m_0)\} \frac{1}{N} \sum_{n=1}^N \mathbb{1} [x_2^{(n)} = x_0^{(n)}] \\ &= \{Q_\epsilon(m_1 | m_0) - Q_\epsilon(m_2 | m_0)\} \left\{ \frac{1}{N} \sum_{n=1}^N \mathbb{1} [x_2^{(n)} = x_0^{(n)}] - \frac{1}{N} \sum_{n=1}^N \mathbb{1} [x_1^{(n)} = x_0^{(n)}] \right\} \\ &= \{Q_\epsilon(m_1 | m_0) - Q_\epsilon(m_2 | m_0)\} \left\{ \left(1 - \frac{\rho(x_0, x_2)}{N}\right) - \left(1 - \frac{\rho(x_0, x_1)}{N}\right) \right\} \\ &= \underbrace{\{Q_\epsilon(m_1 | m_0) - Q_\epsilon(m_2 | m_0)\}}_{>0} \frac{1}{N} \underbrace{\{\rho(x_0, x_1) - \rho(x_0, x_2)\}}_{>0} > 0 \end{aligned}$$

(IV) Payoff upper bound is attained if and only if object distances are preserved

As the swap procedure can strictly increase the expected payoff and the sets of one-to-one languages and objects are finite, repeating this procedure would eliminate all scrambled associations in the language. Since the number of messages that are at distance c from m_0 is equal to the number of objects that are at distance c from x_0 , absence of such a mapping is equivalent to the condition that

$$\rho(x_0, x) = \rho(\ell(x_0), \ell(x)) \quad \text{for all } x \in \mathcal{X}^N,$$

and under this condition the payoff is given by

$$\begin{aligned}
 J(\ell, Q_\epsilon, x_0) &= \sum_{c=0}^N \binom{N}{c} (|\mathcal{V}| - 1)^c (1 - \epsilon)^{N-c} \epsilon^c \left(\frac{1}{|\mathcal{V}| - 1} \right)^c \left(1 - \frac{c}{N} \right) \\
 &= \sum_{c=0}^N \binom{N}{c} (1 - \epsilon)^{N-c} \epsilon^c \left(1 - \frac{c}{N} \right) \\
 &= \sum_{c=0}^N \binom{N}{c} (1 - \epsilon)^{N-c} \epsilon^c - \frac{1}{N} \sum_{c=0}^N \binom{N}{c} (1 - \epsilon)^{N-c} \epsilon^c c \\
 &= 1 - \epsilon \\
 &= 1 - \frac{k}{N}.
 \end{aligned}$$

□

Proof of Proposition 7.1. (I) Payoff of compositional languages

Fix a compositional language ℓ^* , an object $x_0 \in \mathcal{X}^N$, and denote $m_0 = \ell^*(x_0)$. By Lemma A.1, compositionality ensures that for any $m' \in \mathcal{V}^L$ and $x' = \ell^{*-1}(m')$,

$$\rho(m_0, m') = c \iff \rho(x_0, x') = c.$$

Since any admissible corruption scheme $Q \in \mathcal{Q}_k$ satisfies the per-message modification constraint

$$\mathbb{E}_{m' \sim Q(\cdot | m)} \rho(m, m') = k,$$

the average number of modified attributes of x_0 under Q is written as

$$\mathbb{E}_{m' \sim Q(\cdot | m_0)} \rho(m_0, m') = \mathbb{E}_{m' \sim Q(\cdot | m_0)} \rho(x_0, x') = k.$$

Therefore, the expected per-attribute accuracy payoff achieved by any compositional language ℓ^* under any corruption scheme $Q \in \mathcal{Q}_k$ is

$$\begin{aligned}
 J(\ell^*, Q) &= \mathbb{E}_{x \sim \text{Unif}(\mathcal{X}^N)} \mathbb{E}_{m' \sim Q(\cdot | m)} \frac{1}{N} \sum_{n=1}^N \mathbb{I}[x'_n = x_n] \\
 &= \mathbb{E}_{x \sim \text{Unif}(\mathcal{X}^N)} \mathbb{E}_{m' \sim Q(\cdot | m)} \left[1 - \frac{\rho(x, x')}{N} \right] \\
 &= 1 - \frac{k}{N}.
 \end{aligned}$$

Hence, the worst-case payoff of any compositional language ℓ^* is

$$\min_{Q \in \mathcal{Q}_k} J(\ell^*, Q) = 1 - \frac{k}{N}.$$

(II) Payoff upper bound of non-compositional languages

We verify that random noise corruption scheme with an appropriate noise level, Q_ϵ , belongs to $\mathcal{Q}_k$, then show that the payoff of any non-compositional language under Q_ϵ is strictly less than that of compositional languages. We take a similar approach to that of Kuciński et al. (2021). We provide the whole proof for completeness.

Consider a corruption scheme Q_ϵ that modifies each symbol independently with probability ϵ , replacing it with another symbol drawn uniformly at random from the remaining $|\mathcal{V}| - 1$ symbols.

$$Q_\epsilon(m' | m) = (1 - \epsilon)^{L - \rho(m', m)} \epsilon^{\rho(m', m)} \left(\frac{1}{|\mathcal{V}| - 1} \right)^{\rho(m', m)}.$$

Note that Q_ϵ satisfies the per-message modification constraint by the assumption $\epsilon = \frac{k}{L}$, hence $Q_\epsilon \in \mathcal{Q}_k$.

$$\mathbb{E}_{m' \sim Q_\epsilon(\cdot|m)} \rho(m, m') = \sum_{c=0}^L c \binom{L}{c} (1-\epsilon)^{L-c} \epsilon^c = L\epsilon = k.$$

Fix a non-compositional language $\tilde{\ell}$. Since $\tilde{\ell}$ is not compositional, by Lemma A.2, there exist $x_0, x_1, x_2 \in \mathcal{X}^N$ and their corresponding messages $m_i = \tilde{\ell}(x_i)$, such that

$$\rho(x_0, x_1) = \rho(m_0, m_2) = c > 0 \quad \text{and} \quad \rho(x_0, x_2) \neq c, \quad \rho(m_0, m_1) \neq c.$$

Since $\rho(x_0, x_1) = c$ and $\rho(m_0, m_1) \neq c$, $\tilde{\ell}$ does not satisfy the distance preservation condition

$$\rho(x_0, x) = \rho(\tilde{\ell}(x_0), \tilde{\ell}(x)) \quad \text{for all } x \in \mathcal{X}^N$$

On the other hand, any compositional language ℓ^* satisfies this condition by Lemma A.1. Hence, by Lemma A.5, we have

$$J(\tilde{\ell}, Q_\epsilon, x_0) < J(\ell^*, Q_\epsilon, x_0) = 1 - \frac{k}{N}.$$

Thus, there exists an object x_0 such that $J(\tilde{\ell}, Q_\epsilon, x_0) < 1 - \frac{k}{N}$. Since Lemma A.5 guarantees that $J(\ell, Q_\epsilon, x) \leq 1 - \frac{k}{N}$ for all $x \in \mathcal{X}^N$ and $\ell \in \mathcal{L}$, the expectation over $x \sim \text{Unif}(\mathcal{X}^N)$ therefore yields

$$\mathbb{E}_{x \sim \text{Unif}(\mathcal{X}^N)} [J(\tilde{\ell}, Q_\epsilon, x)] < \mathbb{E}_{x \sim \text{Unif}(\mathcal{X}^N)} [J(\ell^*, Q_\epsilon, x)],$$

which is equivalent to

$$J(\tilde{\ell}, Q_\epsilon) < J(\ell^*, Q_\epsilon).$$

(III) *Combining results from (I) and (II)*

For any non-compositional language $\tilde{\ell}$ and compositional language ℓ^* , we have

$$\min_{Q \in \mathcal{Q}_k} J(\tilde{\ell}, Q) \leq J(\tilde{\ell}, Q_\epsilon) < J(\ell^*, Q_\epsilon) = 1 - \frac{k}{N} = \min_{Q \in \mathcal{Q}_k} J(\ell^*, Q).$$

□

Proof of Proposition 7.2. (I) Decomposition of expected payoff into language usage

Fix $w \in \{u, v\}$ and let $w^* = \max_{\ell \in \mathcal{L}} w(\ell)$. Note that for any non-compositional language $\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}$, we have $w(\ell) \leq w^* - \Delta_w$, as Propositions 6.6 and 7.1 state that compositional languages achieve the maximum payoff for u and v . The expected payoff at time t is decomposed as

$$\begin{aligned} \sum_{\ell \in \mathcal{L}} \mathbb{P}_w^{(t)}(\ell) w(\ell) &= \sum_{\ell \in \mathcal{L}_{\text{comp}}} \mathbb{P}_w^{(t)}(\ell) w(\ell) + \sum_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} \mathbb{P}_w^{(t)}(\ell) w(\ell) \\ &\leq \sum_{\ell \in \mathcal{L}_{\text{comp}}} \mathbb{P}_w^{(t)}(\ell) w^* + \sum_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} \mathbb{P}_w^{(t)}(\ell) (w^* - \Delta_w) \\ &= w^* \sum_{\ell \in \mathcal{L}} \mathbb{P}_w^{(t)}(\ell) - \Delta_w \sum_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} \mathbb{P}_w^{(t)}(\ell) \\ &= w^* - \Delta_w \sum_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} \mathbb{P}_w^{(t)}(\ell). \end{aligned}$$

(II) *Writing the regret in terms of the compositional language usage*

From the definition of regret, we have

$$\begin{aligned}
 T \max_{\ell \in \mathcal{L}} w(\ell) - \sum_{t=1}^T \sum_{\ell \in \mathcal{L}} \mathbb{P}_w^{(t)}(\ell) w(\ell) &\geq T w^* - \sum_{t=1}^T \left(w^* - \Delta_w \sum_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} \mathbb{P}_w^{(t)}(\ell) \right) \\
 &= T w^* - T w^* + \Delta_w \sum_{t=1}^T \sum_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} \mathbb{P}_w^{(t)}(\ell) \\
 &= \Delta_w \sum_{t=1}^T \left(1 - \sum_{\ell \in \mathcal{L}_{\text{comp}}} \mathbb{P}_w^{(t)}(\ell) \right) \\
 &= \Delta_w T - \Delta_w \sum_{t=1}^T \sum_{\ell \in \mathcal{L}_{\text{comp}}} \mathbb{P}_w^{(t)}(\ell).
 \end{aligned}$$

Rearranging the above inequality yields

$$\begin{aligned}
 \frac{1}{T} \sum_{t=1}^T \sum_{\ell \in \mathcal{L}_{\text{comp}}} \mathbb{P}_w^{(t)}(\ell) &\geq 1 - \frac{1}{\Delta_w T} \left(T \max_{\ell \in \mathcal{L}} w(\ell) - \sum_{t=1}^T \sum_{\ell \in \mathcal{L}} \mathbb{P}_w^{(t)}(\ell) w(\ell) \right) \\
 &\geq 1 - \frac{R_T^{(w)}}{\Delta_w T},
 \end{aligned}$$

where $R_T^{(w)}$ is the regret bound of the learner under w .

Substituting $w = u$ and $w = v$ into the inequality, we obtain lower bounds for compositional language usage rates under u and v :

$$C_T(u) \geq 1 - \frac{\text{Reg}_T^*}{\Delta_u T} \quad \text{and} \quad C_T(v) \geq 1 - \frac{\text{Reg}_T^\epsilon}{\Delta_v T}.$$

(III) Lower bound for the difference of the two compositionality gaps

By Lemma A.4, below holds for any non-compositional language $\tilde{\ell}$:

$$v(\tilde{\ell}) - u(\tilde{\ell}) \geq \frac{3}{4N|\mathcal{X}|^N} \left(\frac{k}{N(|\mathcal{X}| - 1)} \right)^N$$

Rearranging the above inequality yields

$$u(\tilde{\ell}) \leq v(\tilde{\ell}) - \frac{3}{4N|\mathcal{X}|^N} \left(\frac{k}{N(|\mathcal{X}| - 1)} \right)^N$$

Since the lower bound does not depend on the language, the inequality is preserved when the maximum over all non-compositional languages is taken:

$$\max_{\tilde{\ell} \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} u(\tilde{\ell}) \leq \max_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} v(\ell) - \frac{3}{4N|\mathcal{X}|^N} \left(\frac{k}{N(|\mathcal{X}| - 1)} \right)^N.$$

By Lemma A.1, any compositional language ℓ^* decodes message pairs to object pairs with the same Hamming distance. Therefore, for any corruption scheme $Q \in \mathcal{Q}_k$ that has an expected number of modifications k , we have $J(\ell^*, Q) = 1 - \frac{k}{N}$, which implies $u(\ell^*) = v(\ell^*) = 1 - \frac{k}{N}$. Taking the maximum over all compositional languages preserves the equality:

$$\max_{\ell^* \in \mathcal{L}_{\text{comp}}} u(\ell^*) = \max_{\ell^* \in \mathcal{L}_{\text{comp}}} v(\ell^*).$$

Therefore, we have

$$\begin{aligned}
 \Delta_u &= \max_{\ell^* \in \mathcal{L}_{\text{comp}}} u(\ell^*) - \max_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} u(\ell) \\
 &\geq \max_{\ell^* \in \mathcal{L}_{\text{comp}}} u(\ell^*) - \max_{\tilde{\ell} \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} v(\tilde{\ell}) + \frac{3}{4N|\mathcal{X}|^N} \left(\frac{k}{N(|\mathcal{X}| - 1)} \right)^N \\
 &= \max_{\ell^* \in \mathcal{L}_{\text{comp}}} v(\ell^*) - \max_{\tilde{\ell} \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} v(\tilde{\ell}) + \frac{3}{4N|\mathcal{X}|^N} \left(\frac{k}{N(|\mathcal{X}| - 1)} \right)^N \\
 &= \Delta_v + \frac{3}{4N|\mathcal{X}|^N} \left(\frac{k}{N(|\mathcal{X}| - 1)} \right)^N.
 \end{aligned}$$

Hence,

$$\Delta_u - \Delta_v \geq \frac{3}{4N|\mathcal{X}|^N} \left(\frac{k}{N(|\mathcal{X}| - 1)} \right)^N > 0.$$

Thus, a strictly tighter lower bound for $C_T(u)$ than for $C_T(v)$ is obtained whenever $\text{Reg}_T^* \leq \text{Reg}_T^\epsilon$ with $\text{Reg}_T^\epsilon > 0$:

$$1 - \frac{\text{Reg}_T^*}{\Delta_u T} > 1 - \frac{\text{Reg}_T^\epsilon}{\Delta_v T}.$$

□

Corollary A.6. Suppose $|\mathcal{X}| = |\mathcal{V}| \geq 2$, $N = L \geq 2$. Let $\mathcal{Q}_k$ denote the set of all corruption schemes with expected modification budget $k \in (0, \frac{|\mathcal{V}|-1}{|\mathcal{V}|}L)$:

$$\mathcal{Q}_k = \left\{ Q : \mathbb{E}_{m' \sim Q(\cdot|m)} \rho(m, m') = k, \forall m \in \mathcal{V}^L \right\}.$$

Let $\mathcal{L}$ denote the set of one-to-one languages that map $\mathcal{X}^N$ to $\mathcal{V}^L$, and let $\mathcal{L}_{\text{comp}}$ denote the set of compositional languages in $\mathcal{L}$. Let u and v be the expected payoffs under misinformation and random noise in Definition 6.4 with $\epsilon = \frac{k}{L}$:

$$u(\ell) := \min_{Q \in \mathcal{Q}_k} J(\ell, Q) \quad \text{and} \quad v(\ell) := J(\ell, Q_\epsilon).$$

Let $\delta_{\ell, w} = \max_{\ell' \in \mathcal{L}} w(\ell') - w(\ell)$ be the gap between the optimal payoff and the payoff of language ℓ under $w \in \{u, v\}$, where $\ell \in \mathcal{L}$. For the regret bound of UCB1 under w ,

$$B(w) = 8 \ln T \sum_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} \frac{1}{\delta_{\ell, w}} + \left(1 + \frac{\pi^2}{3} \right) \sum_{\ell \in \mathcal{L}} \delta_{\ell, w},$$

we have $B(u) < B(v)$ when $T \geq 2$.

Proof of Corollary A.6. By Lemma A.1, any pair of messages that are at distance- k is mapped to distance- k objects by a compositional language $\ell^* \in \mathcal{L}_{\text{comp}}$, so it achieves a payoff of $1 - \frac{k}{N}$ under any corruption scheme $Q \in \mathcal{Q}_k$. Proposition 7.1 and Proposition 6.6 state that compositional languages obtain the optimal payoff for both random noise and misinformation, hence the highest payoff is achieved by compositional languages and the corresponding payoffs are equal to each other, $u(\ell^*) = v(\ell^*) = 1 - \frac{k}{N}$, for any compositional language $\ell^* \in \mathcal{L}_{\text{comp}}$.

By Lemma A.4, we have $u(\tilde{\ell}) < v(\tilde{\ell})$ for any non-compositional language $\tilde{\ell} \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}$. Therefore, $\delta_{\tilde{\ell}, u} > \delta_{\tilde{\ell}, v}$ for any

non-compositional language $\tilde{\ell} \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}$. Since $u(\ell^*) = v(\ell^*)$ is the optimal payoff for any compositional language ℓ^* ,

$$\begin{aligned}
 B(u) - B(v) &= 8 \ln T \sum_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} \left(\frac{1}{\delta_{\ell,u}} - \frac{1}{\delta_{\ell,v}} \right) + \left(1 + \frac{\pi^2}{3} \right) \sum_{\ell \in \mathcal{L}} (\delta_{\ell,u} - \delta_{\ell,v}) \\
 &= 8 \ln T \sum_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} \left(\frac{1}{\delta_{\ell,u}} - \frac{1}{\delta_{\ell,v}} \right) + \left(1 + \frac{\pi^2}{3} \right) \sum_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} (\delta_{\ell,u} - \delta_{\ell,v}) \\
 &= \sum_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} \left[8 \ln T \left(\frac{1}{\delta_{\ell,u}} - \frac{1}{\delta_{\ell,v}} \right) + \left(1 + \frac{\pi^2}{3} \right) (\delta_{\ell,u} - \delta_{\ell,v}) \right] \\
 &= \sum_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} \left[8 \ln T \left(\frac{\delta_{\ell,v}}{\delta_{\ell,v}\delta_{\ell,u}} - \frac{\delta_{\ell,u}}{\delta_{\ell,v}\delta_{\ell,u}} \right) + \left(1 + \frac{\pi^2}{3} \right) (\delta_{\ell,u} - \delta_{\ell,v}) \right] \\
 &= \sum_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} \left[8 \ln T \left(\frac{\delta_{\ell,v} - \delta_{\ell,u}}{\delta_{\ell,v}\delta_{\ell,u}} \right) + \left(1 + \frac{\pi^2}{3} \right) (\delta_{\ell,u} - \delta_{\ell,v}) \right] \\
 &= \sum_{\ell \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}} \left[\left(1 + \frac{\pi^2}{3} \right) - \frac{8 \ln T}{\delta_{\ell,v}\delta_{\ell,u}} \right] (\delta_{\ell,u} - \delta_{\ell,v}).
 \end{aligned}$$

Note that $0 < \delta_{\tilde{\ell},v}\delta_{\tilde{\ell},u} \leq 1$ for any $\tilde{\ell} \in \mathcal{L} \setminus \mathcal{L}_{\text{comp}}$, as payoffs lie in the range of $[0, 1]$, and non-compositional languages have strictly positive gaps. By Lemma A.4, $\delta_{\tilde{\ell},u} - \delta_{\tilde{\ell},v} > 0$ for any non-compositional language $\tilde{\ell}$. This implies that $B(u) - B(v) < 0$ whenever $1 + \frac{\pi^2}{3} < 8 \ln T$. Rearranging the above inequality yields

$$1 + \frac{\pi^2}{3} < 8 \ln T \iff \frac{1}{8} + \frac{\pi^2}{24} < \ln T \iff e^{\frac{1}{8} + \frac{\pi^2}{24}} < T.$$

Since $e^{\frac{1}{8} + \frac{\pi^2}{24}} \approx 1.7 < 2$, misinformation yields a strictly smaller UCB1 upper bound expression compared to random noise whenever $T \geq 2$. $\square$

Lemma A.7. Let $\mathcal{X}$ and $\mathcal{V}$ be finite sets with $|\mathcal{X}^N| = |\mathcal{V}^L|$. Let $S : \mathcal{X}^N \rightarrow \mathcal{X}^N$ and $T : \mathcal{V}^L \rightarrow \mathcal{V}^L$ be permutations. Suppose that every $x \in \mathcal{X}^N$ lies on an S -cycle of length t , and every $m \in \mathcal{V}^L$ lies on a T -cycle of length t . Then there exists a one-to-one language $\ell : \mathcal{X}^N \rightarrow \mathcal{V}^L$ such that $\ell \circ S = T \circ \ell$.

Proof of Lemma A.7. Let $\mathcal{C}_S$ be the set of S -cycles in $\mathcal{X}^N$ and $\mathcal{C}_T$ be the set of T -cycles in $\mathcal{V}^L$. Since every cycle has length t , we have $|\mathcal{C}_S| = \frac{|\mathcal{X}^N|}{t}$ and $|\mathcal{C}_T| = \frac{|\mathcal{V}^L|}{t}$, which implies $|\mathcal{C}_S| = |\mathcal{C}_T|$.

Enumerate the cycles as $\mathcal{C}_S = \{C_1^S, \dots, C_r^S\}$ and $\mathcal{C}_T = \{C_1^T, \dots, C_r^T\}$. For each $i \in \{1, \dots, r\}$, we write

$$\begin{aligned}
 C_i^S &= (x_{i,0}, x_{i,1}, \dots, x_{i,t-1}) \quad \text{where} \quad S(x_{i,j}) = x_{i,(j+1) \bmod t}, \\
 C_i^T &= (m_{i,0}, m_{i,1}, \dots, m_{i,t-1}) \quad \text{where} \quad T(m_{i,j}) = m_{i,(j+1) \bmod t}.
 \end{aligned}$$

We take $\ell : \mathcal{X}^N \rightarrow \mathcal{V}^L$ such that $\ell(x_{i,j}) = m_{i,j}$ for all $i \in \{1, \dots, r\}$ and $j \in \{0, \dots, t-1\}$. Note that ℓ is a one-to-one language by construction. Notice that for all i, j ,

$$(T \circ \ell)(x_{i,j}) = T(m_{i,j}) = m_{i,(j+1) \bmod t} = \ell(x_{i,(j+1) \bmod t}) = (\ell \circ S)(x_{i,j}),$$

which implies that $\ell \circ S = T \circ \ell$. $\square$

Proof of Proposition 8.1. (I) Constructing a budget- k modification scheme Q'

Let $q := |\mathcal{X}| = |\mathcal{V}| \geq 2$.

Choose a q -cycle (cyclic permutation) $\sigma_{\mathcal{V}} : \mathcal{V} \rightarrow \mathcal{V}$. Define a permutation $S_K : \mathcal{V}^N \rightarrow \mathcal{V}^N$ by

$$S_K(m_1, \dots, m_N) := (\sigma_{\mathcal{V}}(m_1), \dots, \sigma_{\mathcal{V}}(m_K), m_{K+1}, \dots, m_N),$$

so that $\rho(S_K(m), m) = K$ for every m . Let Q_K be the deterministic channel

$$Q_K(m' | m) := \begin{cases} 1, & \text{if } m' = S_K(m), \\ 0, & \text{otherwise,} \end{cases}$$

and let Q_I be the identity channel

$$Q_I(m' | m) := \begin{cases} 1, & \text{if } m' = m, \\ 0, & \text{otherwise.} \end{cases}$$

Define the mixture channel

$$Q' := \frac{k}{K} Q_K + \left(1 - \frac{k}{K}\right) Q_I.$$

Note that Q' satisfies the expected modification budget constraint for every m ,

$$\sum_{m'} Q'(m' | m) \rho(m', m) = \frac{k}{K} \sum_{m'} Q_K(m' | m) \rho(m', m) + \left(1 - \frac{k}{K}\right) \cdot 0 = \frac{k}{K} K = k,$$

so $Q' \in \mathcal{Q}_k$.

(II) Constructing two non-compositional languages

Choose a q -cycle $\sigma_{\mathcal{X}} : \mathcal{X} \rightarrow \mathcal{X}$. Define permutations on objects $\mathcal{X}^N$:

$$T_{\text{all}}(x_1, \dots, x_N) := (\sigma_{\mathcal{X}}(x_1), \dots, \sigma_{\mathcal{X}}(x_N)) \quad \text{and} \quad T_{\text{one}}(x_1, \dots, x_N) := (\sigma_{\mathcal{X}}(x_1), x_2, \dots, x_N).$$

Note that T_{all} , T_{one} , and S_K apply a single q -cycle to at least one coordinate. Under each of these permutations, the corresponding object or message returns to its original state for the first time after q applications.

Since $\sigma_{\mathcal{X}}$ and $\sigma_{\mathcal{Y}}$ are q -cycles, once the first affected coordinate is fixed, every other affected coordinate is determined by how many applications of the corresponding cycle are needed to reach it. Refer to these numbers as relative offsets. Note that the relative offsets do not change under applications of S_K , T_{all} , or T_{one} . Therefore, every cycle is determined by the relative offsets of the affected coordinates together with the unaffected coordinates.

Let M be the number of affected coordinates. Note that $M = K$ for S_K , $M = N$ for T_{all} , and $M = 1$ for T_{one} . There are q^{M-1} different relative offset patterns, and q^{N-M} different patterns of unaffected coordinates. Note that, when $M = 1$, there are no relative offsets, so the number of relative offset patterns is $q^{M-1} = q^0 = 1$. Similarly, when $M = N$, the number of unaffected coordinate patterns is $q^{N-M} = q^0 = 1$. Hence the number of cycles is given by $q^{M-1} q^{N-M} = q^{N-1}$. Since the first affected coordinate rotates through all q symbols or values, each cycle has length q , and the cycles are disjoint, as each cycle has unique relative offsets and unique unaffected coordinates. Therefore, the permutations T_{all} , T_{one} , and S_K decompose into disjoint q -cycles.

By Lemma A.7, there exist one-to-one languages $\ell', \tilde{\ell} : \mathcal{X}^N \rightarrow \mathcal{Y}^N$ such that

$$\ell' \circ T_{\text{all}} = S_K \circ \ell', \quad \tilde{\ell} \circ T_{\text{one}} = S_K \circ \tilde{\ell}.$$

Notice that on a T_{all} -cycle, ℓ' maps object pairs differing in all N attributes to message pairs differing in only K symbols, while $\tilde{\ell}$ maps object pairs on a T_{one} -cycle differing in one attribute to message pairs differing in K positions.

More specifically, for ℓ' , pick any x and set $y = T_{\text{all}}(x)$, so $\rho(x, y) = N$, but $\rho(\ell'(x), \ell'(y)) = \rho(\ell'(x), S_K(\ell'(x))) = K \neq N$. Similarly, for $\tilde{\ell}$, taking $y = T_{\text{one}}(x)$ gives $\rho(x, y) = 1$, but $\rho(\tilde{\ell}(x), \tilde{\ell}(y)) = K \geq 2$. Therefore, by Lemma A.1, both languages are not compositional.

(III) Showing ℓ' is more vulnerable to Q' compared to random noise

Note that the expected payoff $J(\ell, Q)$ can be written as

$$J(\ell, Q) = \mathbb{E}_{x \sim \text{Unif}(\mathcal{X}^N)} \mathbb{E}_{m' \sim Q(\cdot | \ell(x))} \left(1 - \frac{1}{N} \rho(x, \ell^{-1}(m'))\right).$$

Under Q_K , the decoded object from input x with ℓ' is

$$\ell'^{-1}(S_K(\ell'(x))) = T_{\text{all}}(x),$$

so $\rho(x, T_{\text{all}}(x)) = N$ for all x and hence $J(\ell', Q_K) = 0$. Under Q_1 , $J(\ell', Q_1) = 1$. Therefore

$$J(\ell', Q') = \frac{k}{K} J(\ell', Q_K) + \left(1 - \frac{k}{K}\right) J(\ell', Q_1) = 1 - \frac{k}{K}.$$

Consider the random noise Q_ϵ . With probability $(1 - \epsilon)^L = (1 - \frac{k}{L})^L = (1 - \frac{k}{N})^N$, the message is unchanged and the payoff is 1, while the payoff is always at least 0. Hence

$$J(\ell', Q_\epsilon) \geq \left(1 - \frac{k}{N}\right)^N.$$

Note that $k > K_{\min} = K \left(1 - \left(1 - \frac{K}{N}\right)^N\right)$, so we have

$$\frac{k}{K} > 1 - \left(1 - \frac{K}{N}\right)^N.$$

Since $k < K$, we get $\left(1 - \frac{k}{N}\right)^N > \left(1 - \frac{K}{N}\right)^N$. Combining these two inequalities gives

$$\frac{k}{K} > 1 - \left(1 - \frac{k}{N}\right)^N.$$

Equivalently,

$$1 - \frac{k}{K} < \left(1 - \frac{k}{N}\right)^N.$$

Together with the lower bound on $J(\ell', Q_\epsilon)$, this yields

$$J(\ell', Q') < J(\ell', Q_\epsilon).$$

(IV) Showing $\tilde{\ell}$ exploits Q' and beats every compositional language

Under Q_K , the decoded object from x with $\tilde{\ell}$ is

$$\tilde{\ell}^{-1}(S_K(\tilde{\ell}(x))) = T_{\text{one}}(x),$$

so we have

$$\rho(x, T_{\text{one}}(x)) = 1 \quad \text{for all } x \in \mathcal{X}^N.$$

Hence $J(\tilde{\ell}, Q_K) = 1 - \frac{1}{N}$. Since $J(\tilde{\ell}, Q_1) = 1$, we have

$$J(\tilde{\ell}, Q') = \frac{k}{K} \left(1 - \frac{1}{N}\right) + \left(1 - \frac{k}{K}\right) = 1 - \frac{k}{KN}.$$

Now let ℓ^* be any compositional language. By Lemma A.1, compositional languages preserve object distances. Hence, for any $x \in \mathcal{X}^N$,

$$\rho(x, \ell^{*-1}(S_K(\ell^*(x)))) = \rho(\ell^*(x), S_K(\ell^*(x))) = K,$$

where the last equality follows from the definition of S_K . This implies $J(\ell^*, Q_K) = 1 - \frac{K}{N}$. Hence

$$J(\ell^*, Q') = \frac{k}{K} \left(1 - \frac{K}{N}\right) + \left(1 - \frac{k}{K}\right) = 1 - \frac{k}{N}.$$

Since $K \geq 2$, we have $\frac{k}{KN} < \frac{k}{N}$, and therefore $J(\tilde{\ell}, Q') > J(\ell^*, Q')$ for every compositional language ℓ^* .

□

B. Agent architectures

Sender architecture The sender takes the object $x \in \mathcal{X}^N$ and encodes it with a linear layer. The resulting representation is then passed to a GRU (Cho et al., 2014) block that has a single recurrent layer and hidden size of 128. The GRU layer recursively processes the object for a total of $L = 10$ steps, each time outputting the next symbol of the message from a probability distribution over the alphabet $\mathcal{V}$ of size 10.

Receiver architecture The receiver recursively processes the message with a single-layer GRU of size 128. The last representation is then relayed to N linear layers, each of which corresponds to one of the N attributes and produces activations of size $|\mathcal{X}|$. Softmax activation function induces N distributions over the $|\mathcal{X}|$ values from these activations.

Malign intermediary architecture Two separate linear layers each process the object and the corresponding message. The two activations are summed in an element-wise manner. This initializes a single-layer GRU of size 128 and it produces a modified message in the same manner as the sender.

C. Experimental details

The entropy regularization term is multiplied by scaling hyperparameters of 0.5 and 2.0 for the sender and malign intermediary, respectively. For REINFORCE updates of the sender and intermediary, average reward baselines are employed that subtract each agent’s historical mean reward from its current reward. The intermediary takes an additional loss corresponding to the deviations from the number of symbols to be modified, k . This is written as $|k - k'|$, where k' denotes the number of modified symbols. All agents are optimized with ADAM (Kingma & Ba, 2015) optimizer. We use a learning rate of 0.001 and set $\beta_1 = 0.9$, $\beta_2 = 0.999$. We train the agents for approximately 183K steps with a batch size of 4096.

Following a similar approach to Chaabouni et al. (2020), we discard a few experiments that fail to achieve training accuracy of 95% at the end of training. Similar to Li & Bowling (2019); Gupta et al. (2020); Slowik et al. (2021); Korbak et al. (2019), we evaluate languages produced by greedy decoding from the sender on the entire set of objects $\mathcal{X}^N$. We also note that no significant differences in compositionality measures are observed across dataset splits. For the generalization metric, we additionally report in Appendix D the compositional generalization ability of languages on the test set that attained the highest validation accuracy during training to measure the peak generalization performance.

D. Complete experimental results

This section presents complete experimental results for all compositionality measures across the experimental conditions discussed in the main text. Generalization (Val. Max) denotes the test set accuracy of the language that attained the highest validation accuracy during training.

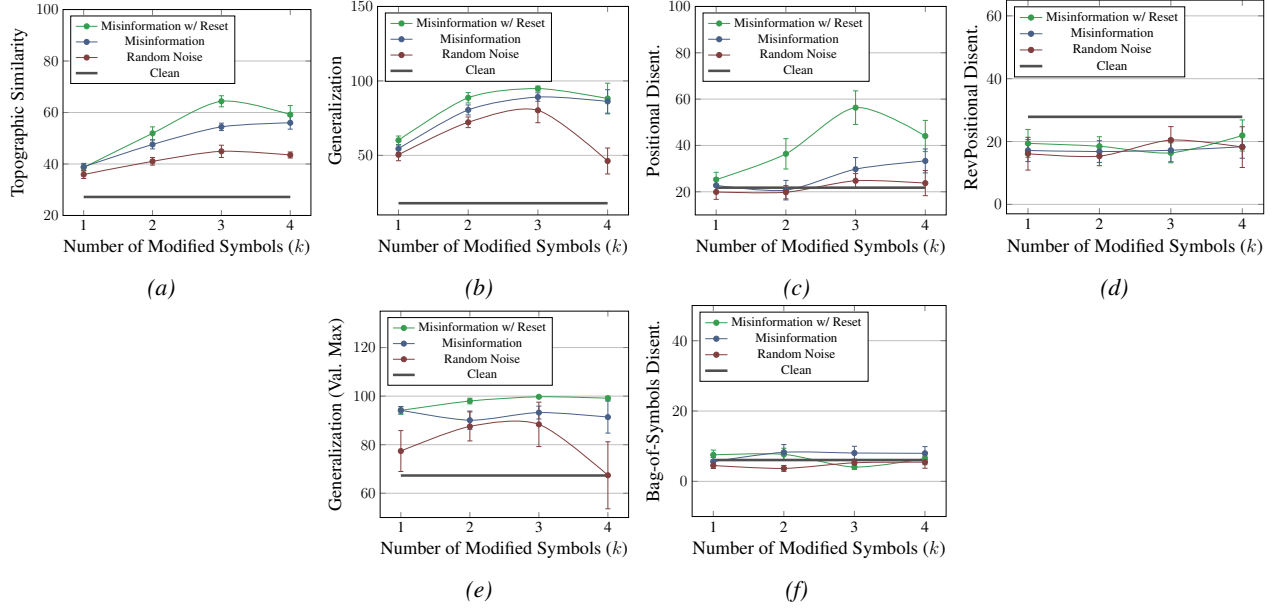


Figure 10. Comparison of language properties under misinformation with reset regime.

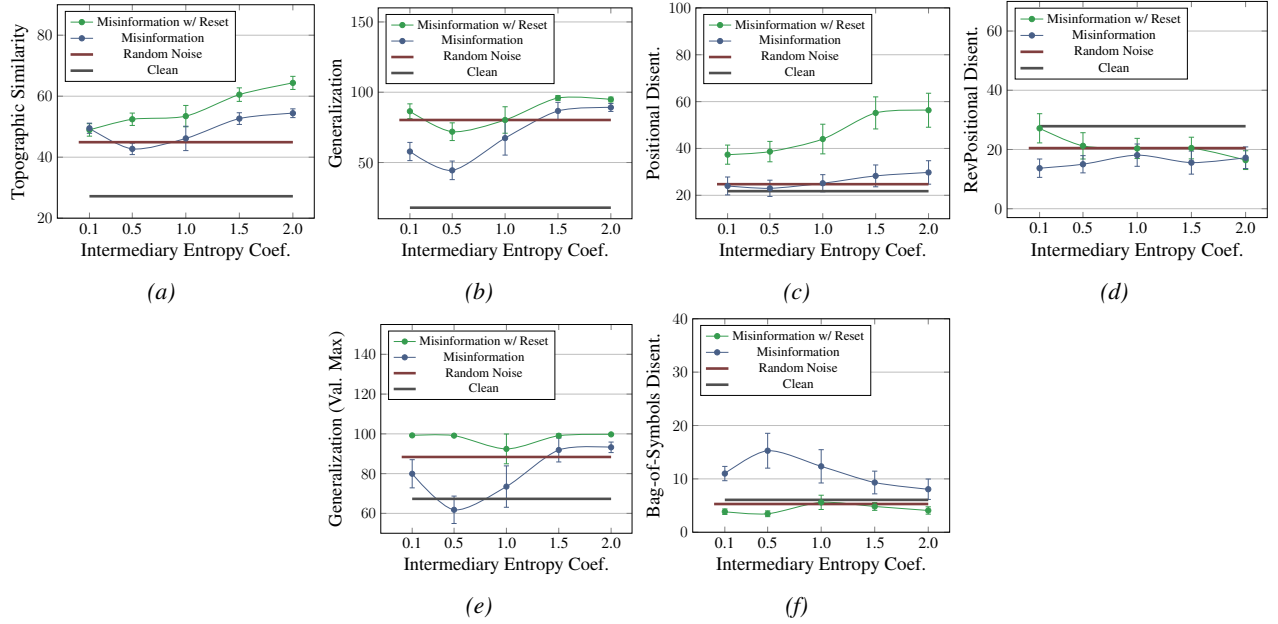


Figure 11. Language properties under varying intermediary entropy coefficients in misinformation with and without reset regimes.

Metric	Condition	(4, 18)	(6, 7)
TopSim	Misinfo w/ Reset	55.2 ±6.8	63.8 ±4.1
	Misinformation	46.6±2.7	55.4±3.4
	Random	46.6±3.5	44.6±3.4
	Clean	18.4±2.4	32.3±6.7
Generalization	Misinfo w/ Reset	86.6±11.0	96.7 ±5.5
	Misinformation	85.6±7.8	87.1±12.6
	Random	89.6 ±4.8	72.9±11.3
	Clean	8.9±7.8	24.0±21.5
PosDis	Misinfo w/ Reset	60.3 ±15.7	47.7 ±8.7
	Misinformation	33.6±9.5	26.7±7.0
	Random	33.4±5.9	19.2±5.6
	Clean	15.4±5.0	20.1±11.7
RevPosDis	Misinfo w/ Reset	9.2 ±4.7	34.2 ±13.9
	Misinformation	8.1±3.0	21.0±6.1
	Random	7.7±3.2	24.3±10.0
	Clean	<u>8.9</u> ±3.2	<u>26.8</u> ±19.6
Gen. (Val. Max)	Misinfo w/ Reset	96.2 ±12.7	99.9 ±0.1
	Misinformation	91.0±6.0	91.9±11.8
	Random	93.3±4.4	88.2±11.7
	Clean	66.7±20.7	75.7±20.3
BosDis	Misinfo w/ Reset	5.7±2.3	4.5±2.5
	Misinformation	10.0 ±5.4	<u>5.7</u> ±2.9
	Random	7.9±2.3	3.5±1.2
	Clean	2.4±1.1	7.3 ±9.2

Table 2. Properties of languages induced from varying attribute-value configurations. **Bold** and underlined numbers indicate the highest and second-highest values in each column. Numbers to the right of $\pm$ denote standard deviations.

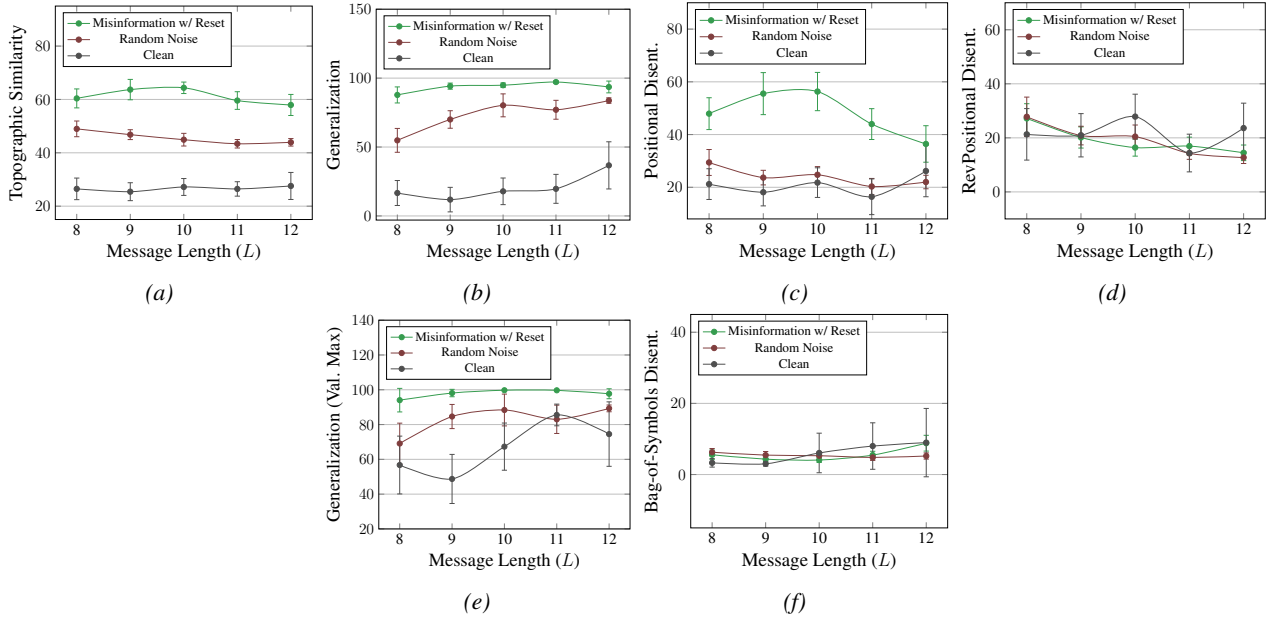


Figure 12. Properties of languages under different message length configurations.

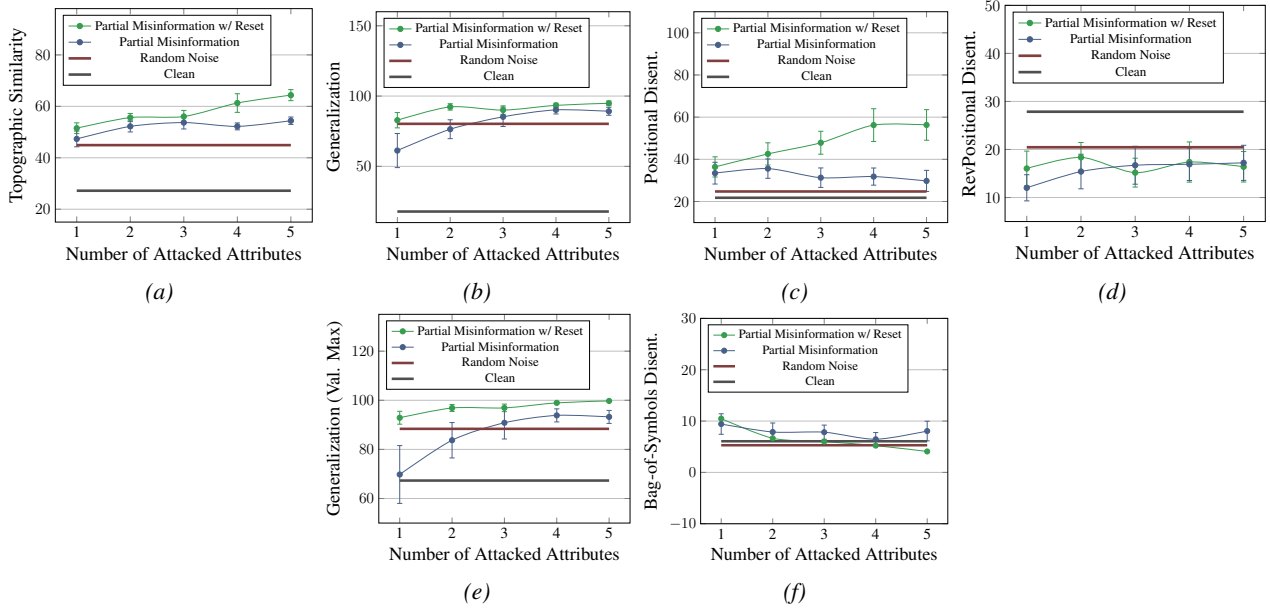


Figure 13. Properties of emergent languages by number of attacked attributes under partial misinformation.

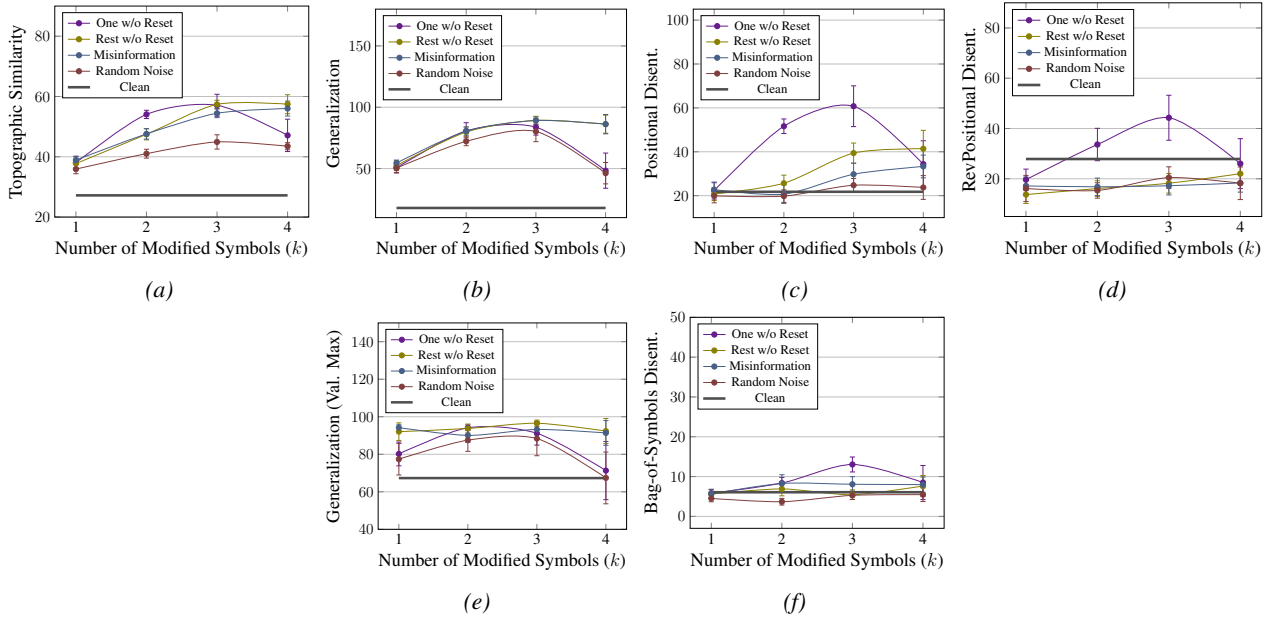


Figure 14. Properties of languages induced under the shifting interest setup in partial misinformation environment without intermediary resets.

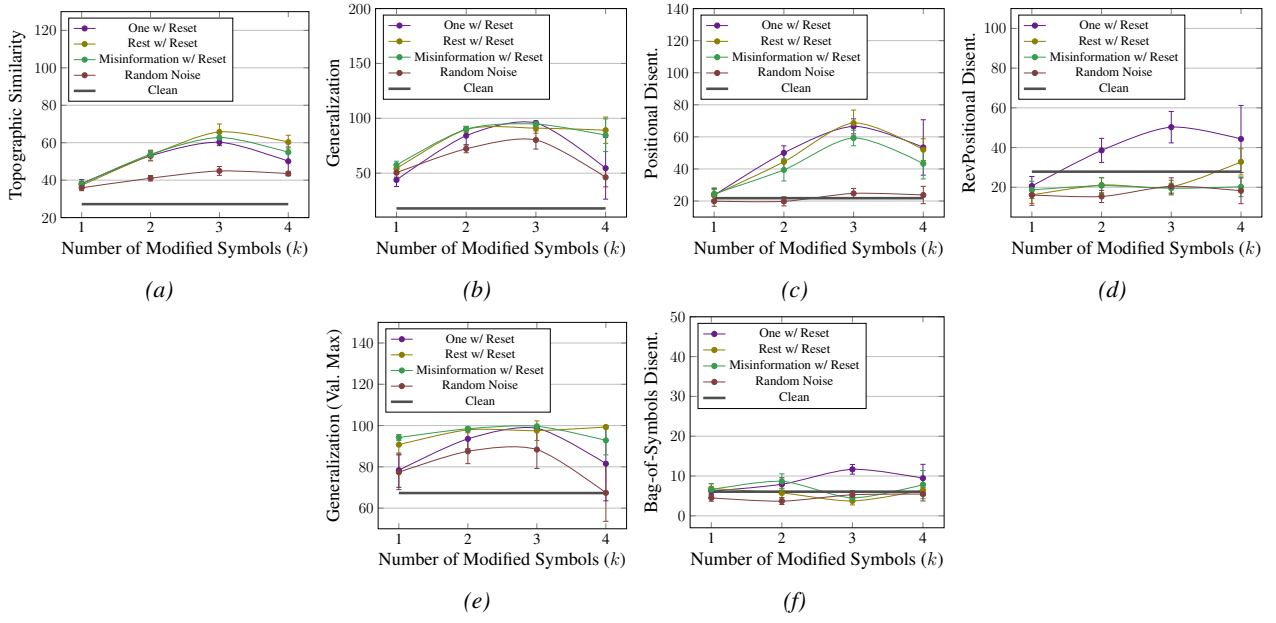


Figure 15. Properties of languages induced under the shifting interest setup in partial misinformation environment with intermediary resets.

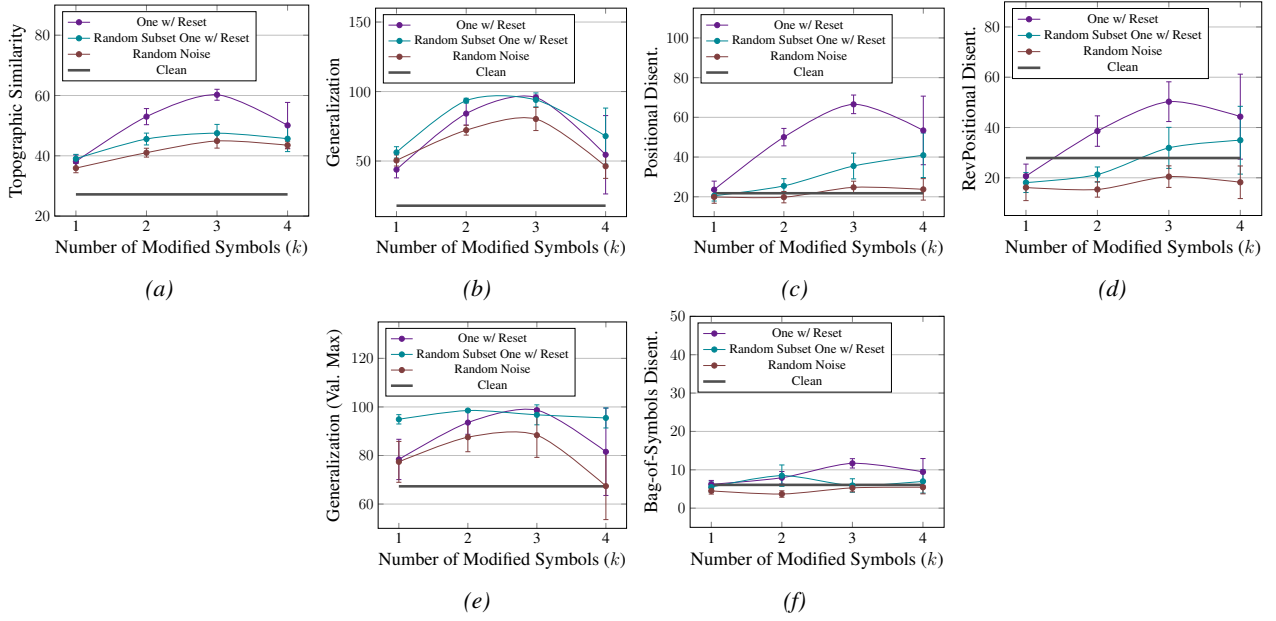


Figure 16. Properties of languages induced under per-step random interests setup in partial misinformation environment.

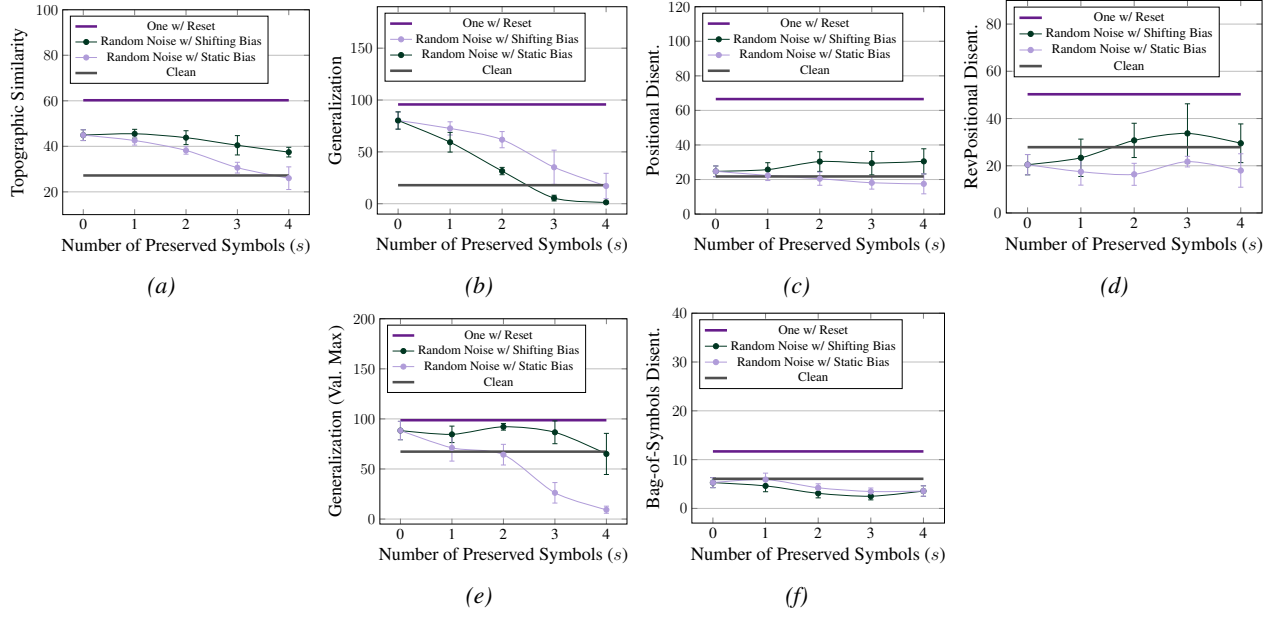


Figure 17. Properties of languages under different numbers of preserved symbols in biased random noise environments.

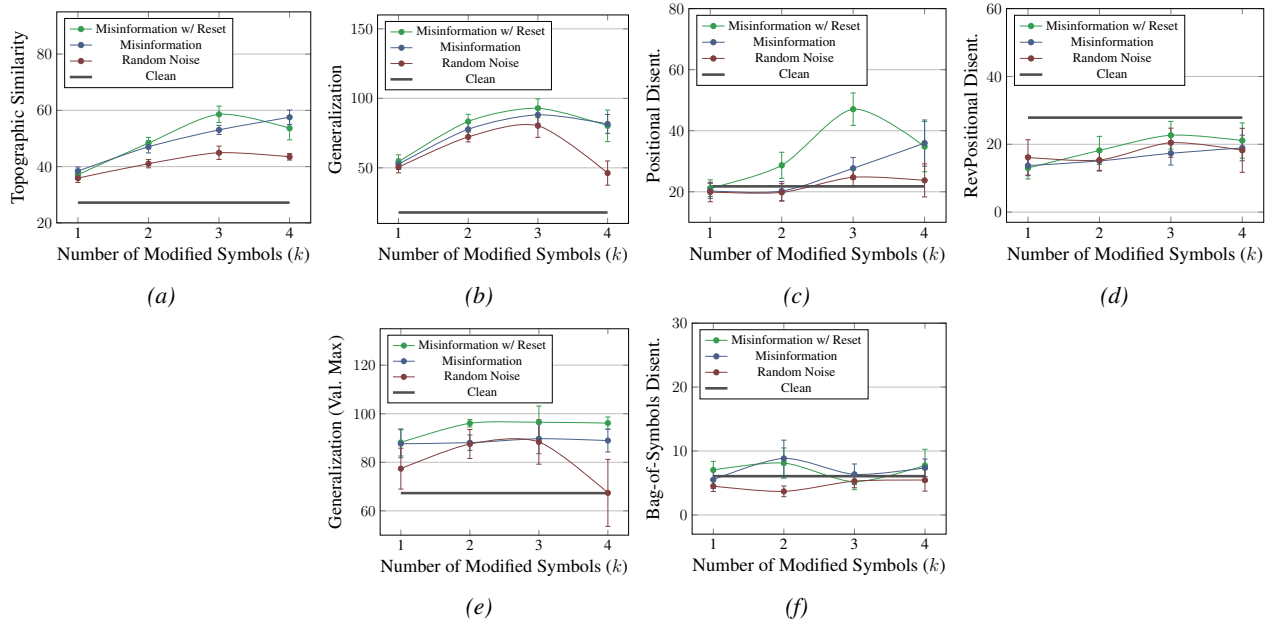


Figure 18. Language properties in the environment where intermediaries can only observe the sender's messages.

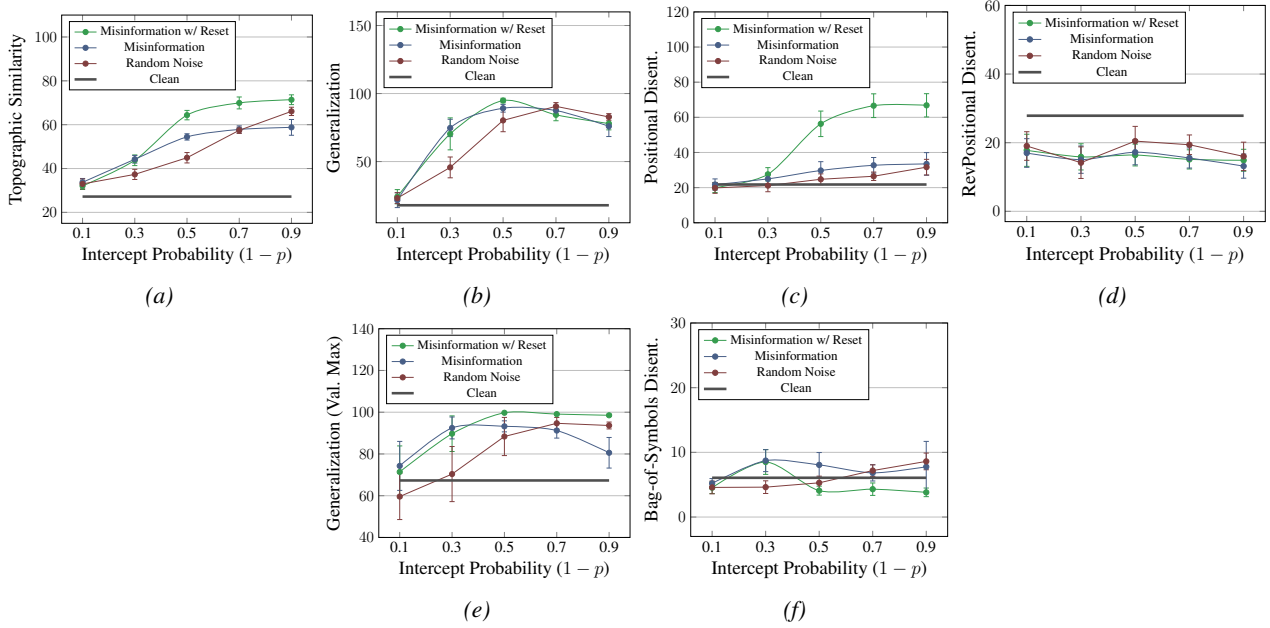


Figure 19. Language properties under varying values of intercept probability ($1 - p$). A message is intercepted by the intermediary with probability $1 - p$. The number of modified symbols is set to $k = 3$. Misinformation w/ Reset exhibits the highest peak scores of TopSim, generalization, and PosDis over intercept probabilities. Relatively high values of TopSim and strong generalization abilities are observed in misinformation (with and without resets) at lower values of intercept probability compared to random noise.

E. Experiments with 3dshapes dataset

3dshapes dataset (Burgess & Kim, 2018) consists of synthetic images of 3D objects. Each image is generated from six independent latent attributes (object shape, object color, floor color, wall color, scale, and orientation). Each attribute has at least four possible values. We use all of the attributes and take four values from each attribute for the experiments. We take 40% of the samples as the train set, 10% as the validation set, and the rest as the test set. We set the alphabet size, $|\mathcal{V}|$, and message length, L , to 8. The agents are trained for 64K steps. In the environment where new intermediaries are periodically introduced, we reset the intermediary’s parameters at eight evenly spaced intervals during training. The sender and intermediary entropy regularization coefficients are set to 0.2 and 2.0, respectively. Other details follow the main experimental setup.

We replace the linear object embedding layer with a CNN encoder to process image inputs in the sender and intermediary architectures. The convolutional layer consists of 3 input channels, 2 output channels, a 3×3 filter, stride 1, and same padding. The output of the convolutional layer is fed to an ELU activation (Clevert et al., 2016) and a max-pooling layer with a 4×4 pooling window and stride 4. The resulting feature map is flattened into a vector of size 512, then projected to size 128 by a linear layer, and then passed through another ELU activation. Other architectural configurations follow the same setup as in the main experiments.

We take the all-attribute setup with intermediary resets described in Section 8 and compare languages induced under misinformation with those induced under random noise. In Figure 20, we observe similar trends to those in the experiments with the simple attribute-value dataset. High degrees of TopSim values are observed under misinformation compared to random noise, and the intermediary reset regime tends to exhibit higher TopSim values than the regime without intermediary reset. Similar patterns are observed in compositional generalization and PosDis scores. BosDis scores tend to remain unaffected under varying degrees of modification strengths, k , consistent with the results from the main experiments. These results suggest that the findings of the main experiments are not limited to the synthetic attribute-value environment and that they may generalize to more complex and realistic environments where attributes are more entangled in the input space, such as images.

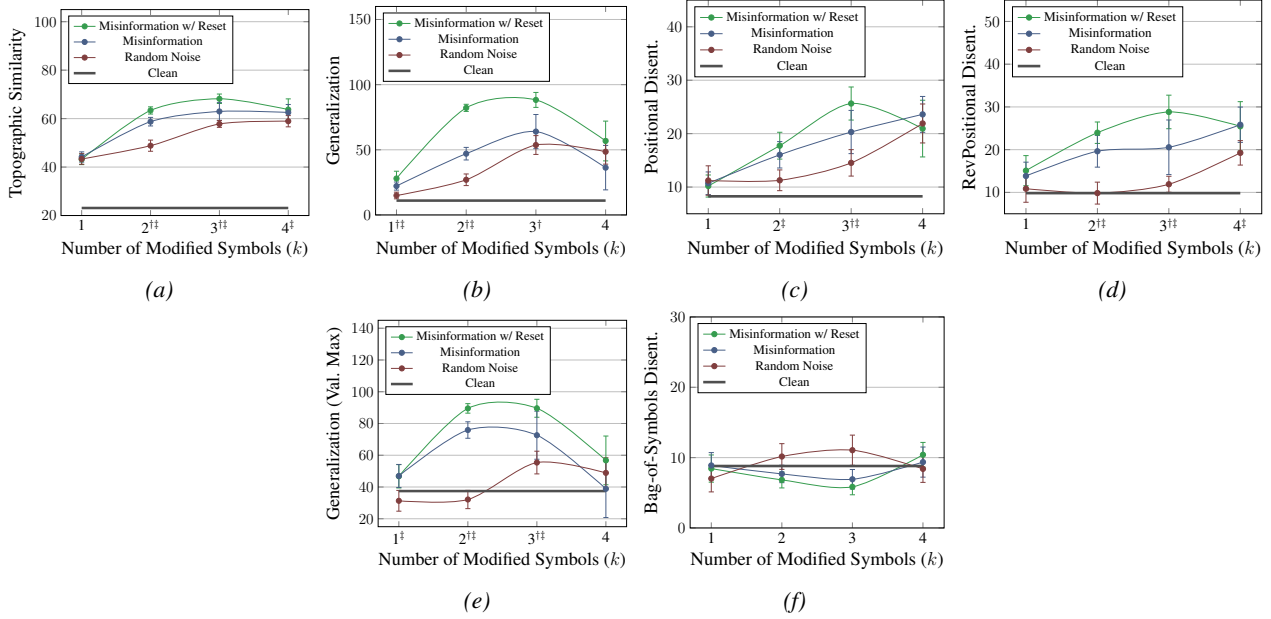


Figure 20. Properties of languages induced on 3dshapes dataset. Statistically significant differences for misinformation with reset relative to misinformation without reset are denoted by [†]. Statistically significant differences for misinformation without reset relative to random noise are denoted by [‡].

F. Analysis of learned corruption schemes

F.1. Communication accuracy on modified messages

The experimental setup employs a post hoc budget correction mechanism in which the intermediary’s corrupted messages are adjusted to ensure that exactly k symbols are modified. This mechanism is designed to maintain a consistent level of corruption across different conditions, allowing for a controlled comparison between misinformation and random noise environments.

This post hoc correction introduces variability in the intermediary’s reward signal, as the intermediary’s reward is calculated based on the modified message after correction. This means that the intermediary’s learning process is influenced by the corrected messages, which may not always reflect the original corruption scheme intended by the intermediary.

However, as the intermediary’s reward is calculated using the budget-corrected message, the post hoc correction can be considered a form of noise. This suggests that just as the sender and receiver can learn to communicate with each other despite the risk of message corruption, the intermediaries should be able to learn appropriate corruption schemes under the influence of budget correction.

To verify that intermediaries successfully learn to corrupt messages under budget correction, we analyze the accuracy of sender-receiver communication on corrupted messages during training. The lower values of accuracy under misinformation in Figure 21 suggest that intermediaries learn meaningful corruption schemes. Misinformation exhibits lower accuracy than random noise. Misinformation with intermediary reset shows spikes when new intermediaries are introduced, then decreases further below the misinformation without reset condition.

F.2. Distribution of modified symbol positions

We analyze which positions the learned corruption schemes of intermediaries are more likely to target. We visualize the proportion of symbol modifications occurring at each position in modified messages during training. Modification proportion at position i measures the frequency of modifications at that position. For example, a value of 0.3 at position i indicates that 30% of all modified symbols occurred at position i after the post hoc budget correction. The number of modified symbols is set to $k = 3$.

In Figure 22, we observe that misinformation tends to modify symbols toward the end of a message. Misinformation

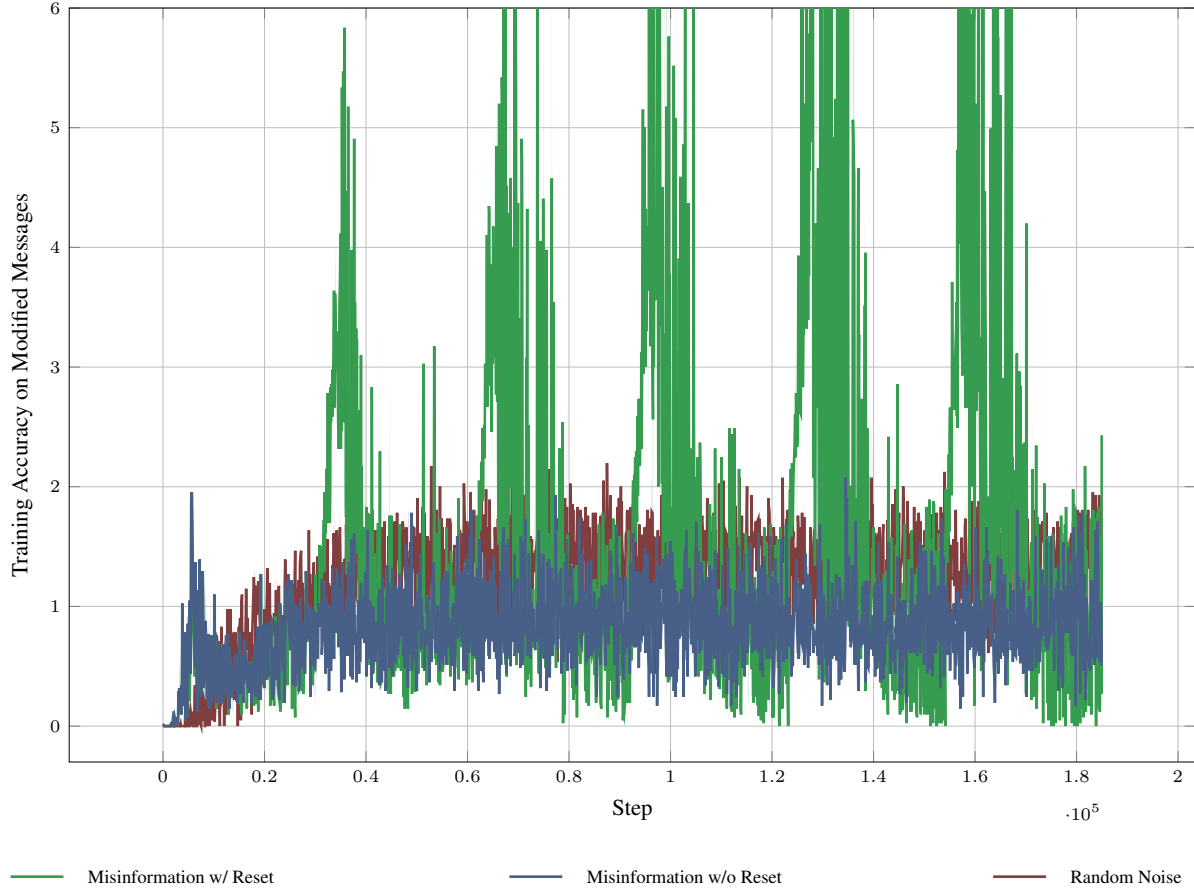


Figure 21. Training accuracy on modified messages under modification budget $k = 4$. Shaded regions denote minimum and maximum values across random seeds.

with the reset regime shows more evenly distributed modification patterns compared to the misinformation without reset setup, suggesting a more adaptive behavior. We cautiously speculate that greedy sampling from the sender may be a factor in producing more corruption-robust phrases at the beginning of a message. However, we refrain from drawing firm conclusions, as further analysis is needed.

We also analyze the entropy of modified proportions of positions during training. In Figure 23, we observe that misinformation learns to apply positionally targeted modifications. New intermediaries maintain a more balanced distribution compared to the misinformation without reset regime, suggesting a higher degree of adaptability.

F.3. Analysis of RevPosDis scores in partial misinformation

We analyze the pre-average attribute-wise RevPosDis term for each attribute a_j , $\frac{\mathcal{I}(a_j; m_1^j) - \mathcal{I}(a_j; m_2^j)}{\mathcal{H}(a_j)}$, during training. High scores indicate that the attribute a_j is uniquely represented by a single position. We take the *one-attribute* setting with intermediary reset setup in the partial misinformation environment in Section 9. In this setup, the intermediary’s target attribute sequentially changes from 1 to 5.

In Figure 24, we observe that the attribute-wise RevPosDis score decreases when the corresponding attribute is targeted by the intermediary. We hypothesize that this may be due to the incentive to spread information across positions for redundancy, gaining protection from modification. For example, representing an attribute over five positions could be more resilient to a single-position corruption than representing the attribute in a single position.

The score then spikes once the intermediary’s target attribute shifts from the corresponding attribute. This suggests that the attribute’s information becomes more concentrated in a single position. We speculate this is due to the process of making

space for the subsequent target attribute, which requires redundant space for protection; the disentangled previous target attribute (see Appendix F.4 for discussions on how a targeted attribute can become disentangled) is then squeezed into a small number of positions, yielding a high attribute-wise RevPosDis score for the corresponding attribute.

F.4. Analysis of PosDis scores in partial misinformation

Figure 25 visualizes the modification proportion under partial misinformation across positions during training. In this setup, the intermediary’s target attribute sequentially changes from 1 to 5 during training at evenly spaced intervals. A modification proportion value of 0.3 at a position indicates that 30% of all modified symbols occurred at that position.

We analyze the pre-average position-wise PosDis term for each position i , $\frac{\mathcal{I}(m_i; a_1^i) - \mathcal{I}(m_i; a_2^i)}{\mathcal{H}(m_i)}$, during training. A high score indicates that the i -th position uniquely conveys information about a single attribute. In Figure 26, we observe that spikes in the modification proportion at a position are often accompanied by spikes in the corresponding position’s position-wise PosDis score. This suggests that targeted attacks may reduce the amount of non-target attribute information in that position. We hypothesize that this is because mixing target and non-target attributes in the same position exposes the non-target information to attacks, whereas positional separation yields safe zones for non-target attributes.

G. Pseudocode for training iteration

Algorithm 1 Training step for communication game under misinformation

Require: object $x = (x_1, \dots, x_N) \in \mathcal{X}^N$, target attribute index set $\mathcal{S} \subseteq \{1, \dots, N\}$, sender π_ϕ , receiver π_θ , intermediary π_ψ , alphabet $\mathcal{V}$, pass-through probability p , modification budget k , sender entropy coefficient λ_ϕ , intermediary entropy coefficient λ_ψ , sender reward baseline b_ϕ , intermediary reward baseline b_ψ

- 1: sample sender message $m \sim \pi_\phi(\cdot | x)$
- 2: sample $c \sim \text{Bernoulli}(1 - p)$
- 3: **if** $c = 0$ **then**
- 4: $m' \leftarrow m$
- 5: $k' \leftarrow 0$
- 6: **else**
- 7: sample modified message from intermediary $\tilde{m} \sim \pi_\psi(\cdot | x, m)$
- 8: $m' \leftarrow \text{BudgetCorrection}(\tilde{m}, m, k)$
- 9: **end if**
- 10: compute receiver reconstruction loss $\ell_\theta \leftarrow -\frac{1}{N} \sum_{n=1}^N \log \pi_\theta(x_n | m')$
- 11: update receiver parameters $\theta \leftarrow \text{OptimizerStep}(\theta, \nabla_\theta \ell_\theta)$
- 12: compute sender entropy $h_\phi \leftarrow -\frac{1}{|m|} \sum_{t=1}^{|m|} \sum_{v \in \mathcal{V}} \pi_\phi(v | x, m_{<t}) \log \pi_\phi(v | x, m_{<t})$
- 13: compute sender loss $\ell_\phi \leftarrow -(\ell_\theta - b_\phi) \log \pi_\phi(m | x) - \lambda_\phi h_\phi$
- 14: update sender parameters $\phi \leftarrow \text{OptimizerStep}(\phi, \nabla_\phi \ell_\phi)$
- 15: **if** $c = 1$ **then**
- 16: compute intermediary entropy $h_\psi \leftarrow -\frac{1}{|\tilde{m}|} \sum_{t=1}^{|\tilde{m}|} \sum_{v \in \mathcal{V}} \pi_\psi(v | x, m, \tilde{m}_{<t}) \log \pi_\psi(v | x, m, \tilde{m}_{<t})$
- 17: compute intermediary modification count $k' \leftarrow \rho(\tilde{m}, m)$
- 18: compute target-index receiver loss $\ell_\mathcal{S} \leftarrow -\frac{1}{|\mathcal{S}|} \sum_{n \in \mathcal{S}} \log \pi_\theta(x_n | m')$
- 19: compute intermediary loss $\ell_\psi \leftarrow -(\ell_\mathcal{S} - b_\psi) \log \pi_\psi(\tilde{m} | x, m) - \lambda_\psi h_\psi + |k - k'|$
- 20: update intermediary parameters $\psi \leftarrow \text{OptimizerStep}(\psi, \nabla_\psi \ell_\psi)$
- 21: **end if**
- 22: **function** $\text{BudgetCorrection}(\tilde{m}, m, k)$
- 23: initialize corrected message $m' \leftarrow \tilde{m}$
- 24: identify modified positions $\mathcal{I}_{\text{modified}} \leftarrow \{i \in \{1, \dots, |\tilde{m}|\} : \tilde{m}_i \neq m_i\}$
- 25: identify unmodified positions $\mathcal{I}_{\text{unmodified}} \leftarrow \{i \in \{1, \dots, |\tilde{m}|\} : \tilde{m}_i = m_i\}$
- 26: **if** $|\mathcal{I}_{\text{modified}}| > k$ **then**
- 27: sample positions to revert $\mathcal{I}_{\text{revert}} \subset \mathcal{I}_{\text{modified}}$ with $|\mathcal{I}_{\text{revert}}| = |\mathcal{I}_{\text{modified}}| - k$
- 28: **for** $i \in \mathcal{I}_{\text{revert}}$ **do**
- 29: revert selected position $m'_i \leftarrow m_i$
- 30: **end for**
- 31: **else if** $|\mathcal{I}_{\text{modified}}| < k$ **then**
- 32: sample positions to modify $\mathcal{I}_{\text{modify}} \subset \mathcal{I}_{\text{unmodified}}$ with $|\mathcal{I}_{\text{modify}}| = k - |\mathcal{I}_{\text{modified}}|$
- 33: **for** $i \in \mathcal{I}_{\text{modify}}$ **do**
- 34: replace selected position $m'_i \sim \text{Unif}(\mathcal{V} \setminus \{m_i\})$
- 35: **end for**
- 36: **end if**
- 37: **return** m'
- 38: **end function**

Random noise and clean channel variants: Random noise replaces the intermediary block with uniform substitutions at exactly k positions. The clean channel sets $m' \leftarrow m$.

H. Reproducibility

The code used to conduct the experiments in this work is available at

- <https://github.com/hy18284/misinformation>
- <https://github.com/MLAI-Yonsei/misinformation>

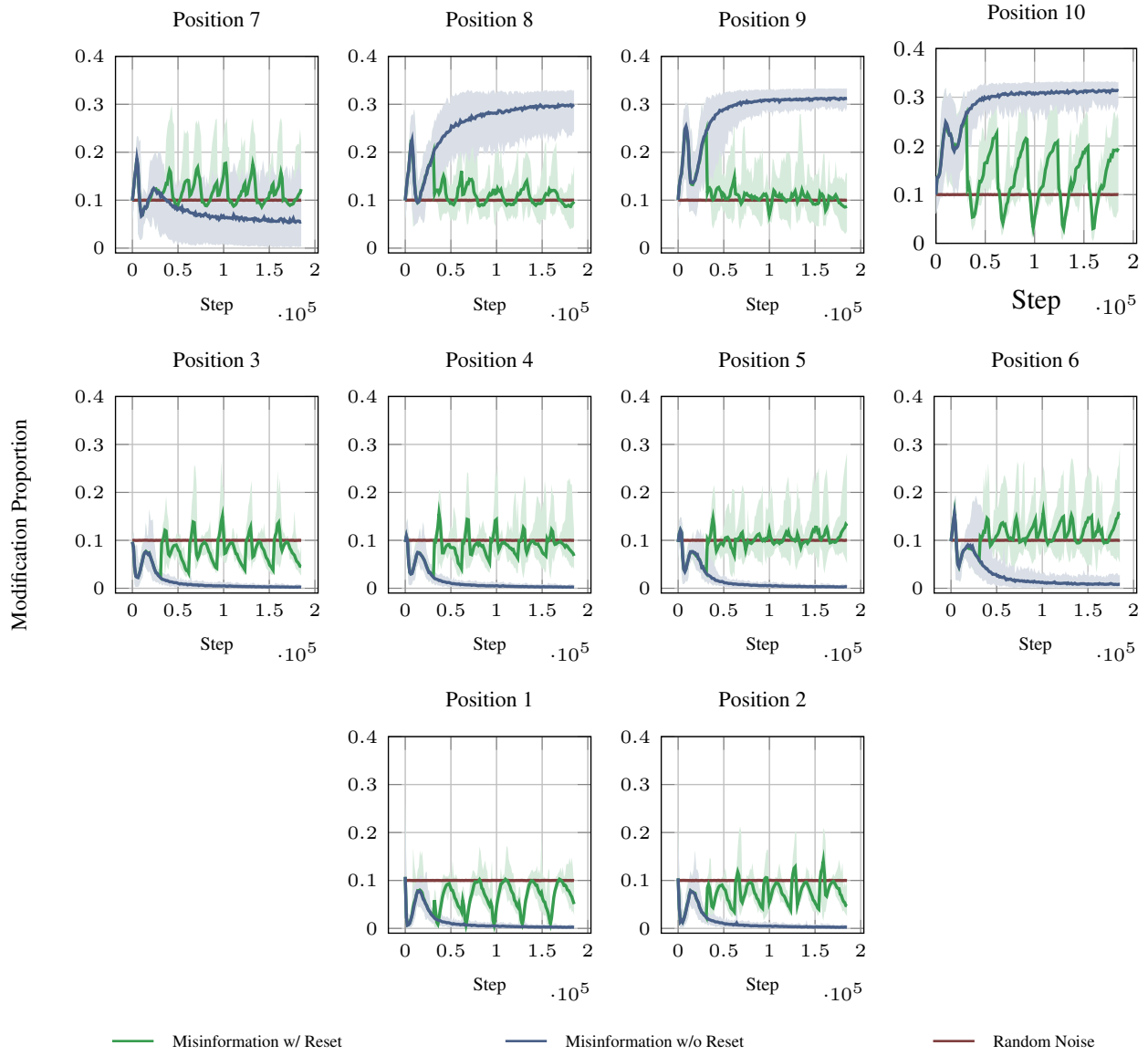


Figure 22. Modification proportion of each position in modified messages during training. Shaded regions denote minimum and maximum values across random seeds.

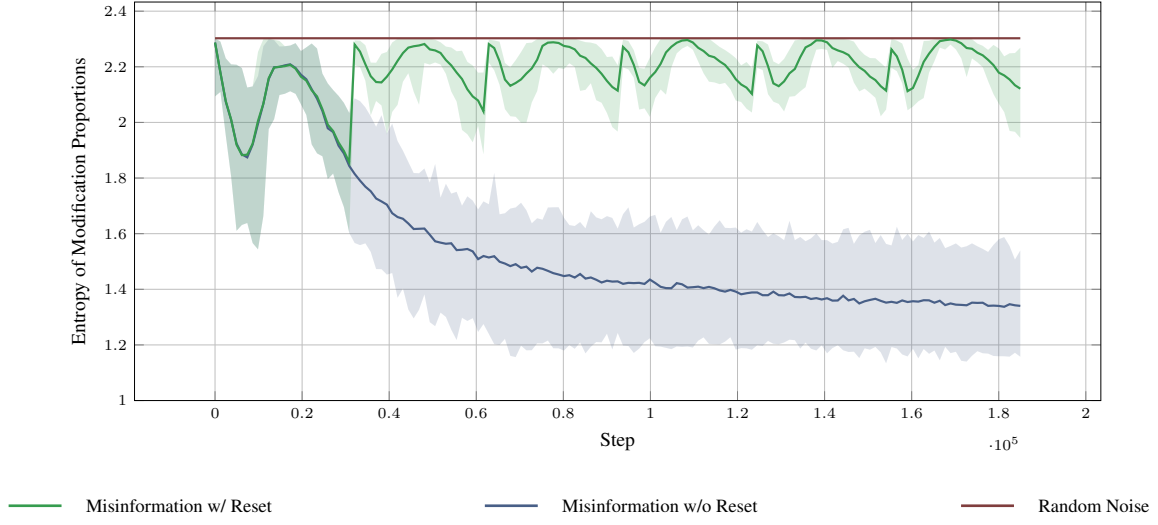


Figure 23. Entropy of modified proportions of positions during training. Shaded regions denote minimum and maximum values across random seeds.

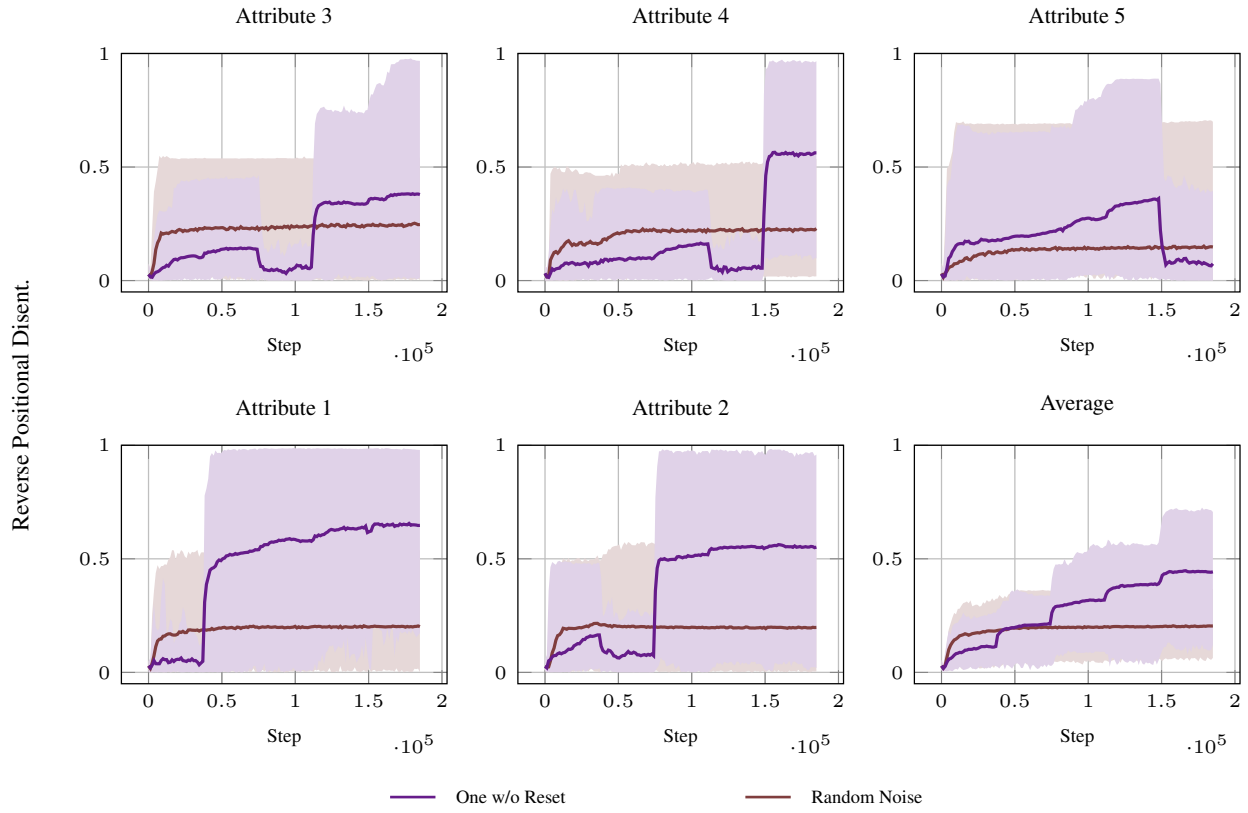


Figure 24. Attribute-wise RevPosDis scores across five attributes during training. Average denotes the average score over all attributes, corresponding to the RevPosDis score defined in Section 5. Shaded regions denote minimum and maximum values across random seeds.

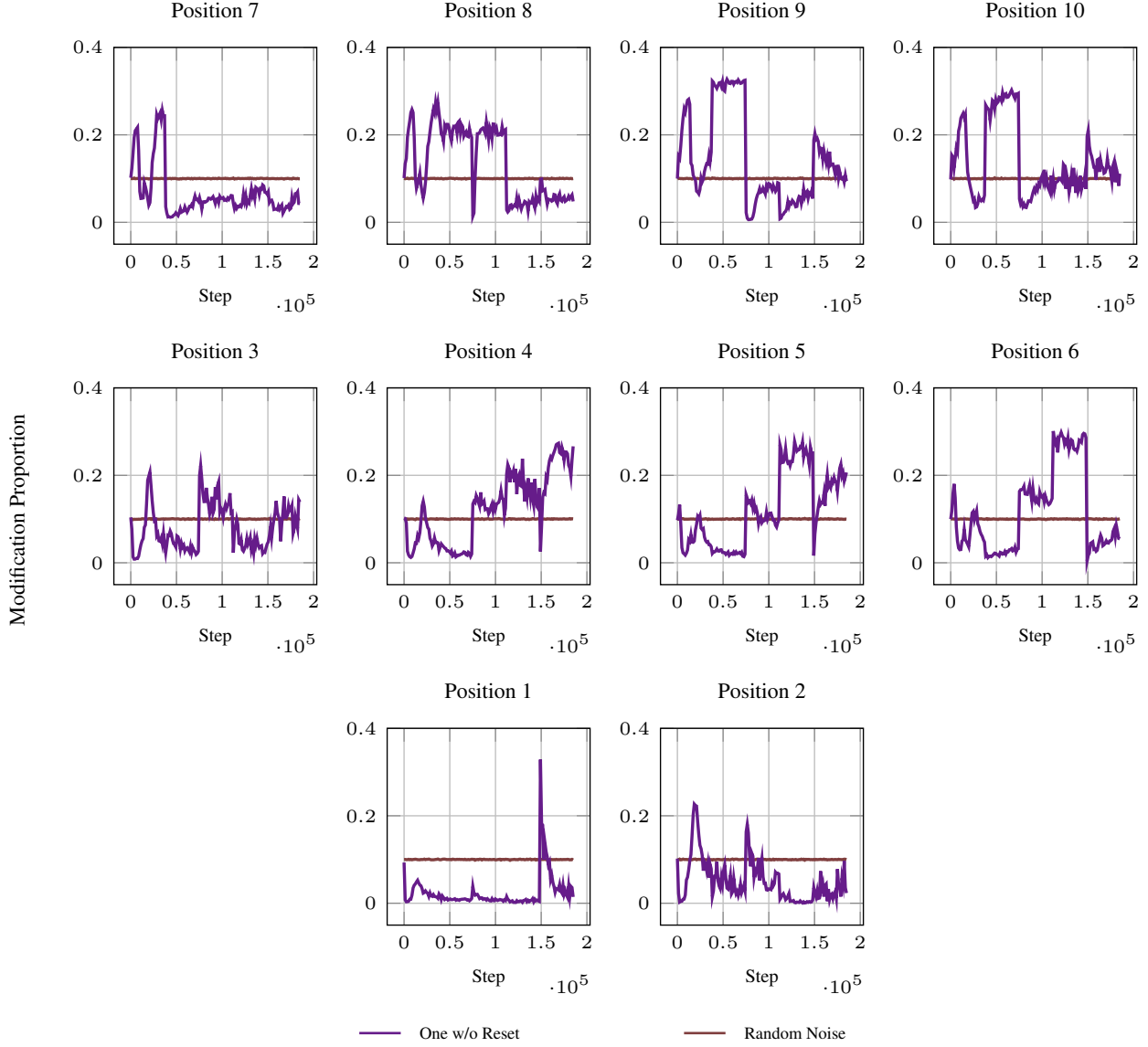


Figure 25. Modification proportion under partial misinformation across ten message positions. A single-seed experiment is shown, as the position used to represent a given attribute is arbitrary across experiments. The modification budget is set to $k = 3$.

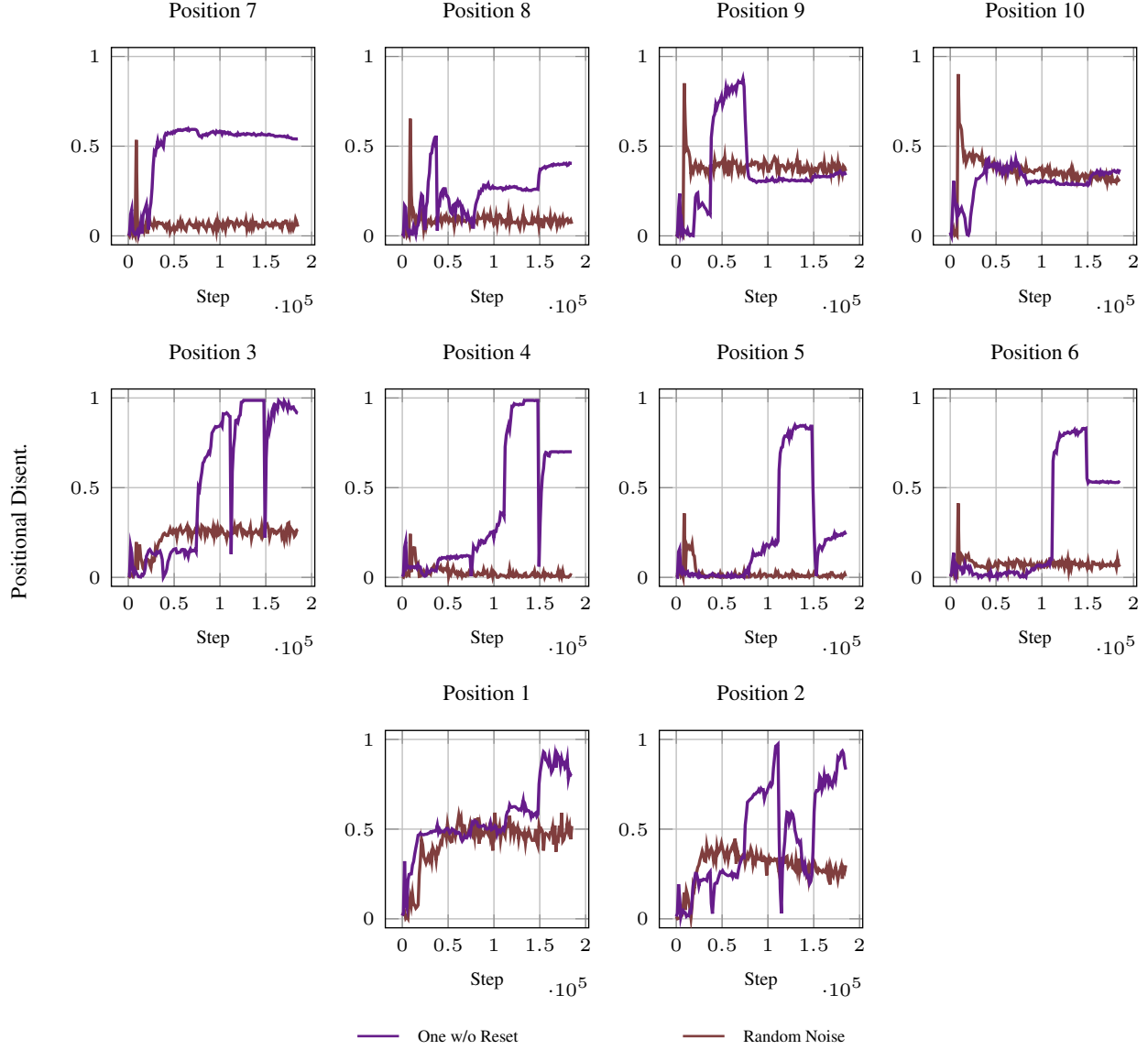


Figure 26. Position-wise PosDis scores in partial misinformation across ten message positions during training. A single-seed experiment is shown, as the position used to represent a given attribute is arbitrary across experiments. The modification budget is set to $k = 3$.